

AKADEMIKA

DCL- DPCM

**DPCM / ADPCM
MODULATION / DEMODULATION
KIT**

EXPERIMENTAL MANUAL

....A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ADVANCED DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO:

-Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to the packing list. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In DCL-DPCM kit, signals are to be monitored with an oscilloscope as explained in the manual. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

These kits must be employed only for the use for which it has been conceived, i.e. as educational equipment, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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TECHNICAL SPECIFICATIONS

Onboard Signals

Sine Wave

Frequency : 500Hz
Amplitude : 0 to 4Vpp

Data Clock

Frequencies : 64 KHz, 128 KHz, 256 KHz and 512 KHz
Amplitude : 5V

Sampling Clock

Frequency : 16 KHz
Amplitude : 5V

ADC Word Length : 8 bits

Modulation Techniques : DPCM using sampler quantize and linear predictor.

Data Clock : 64 KHz to 512 KHz

ADPCM using CODEC

Sync: 8 KHz

Enable: 8 KHz

Data Clock: 64 KHz, 256 KHz, 512 KHz

Channel : Single channel

Channel Operation : Simplex

Coding : μ -law

Switch Faults : 8 nos.

Interconnections : 2mm banana sockets

Test Points : Several test points are provided onboard to
Observe intermediate signals

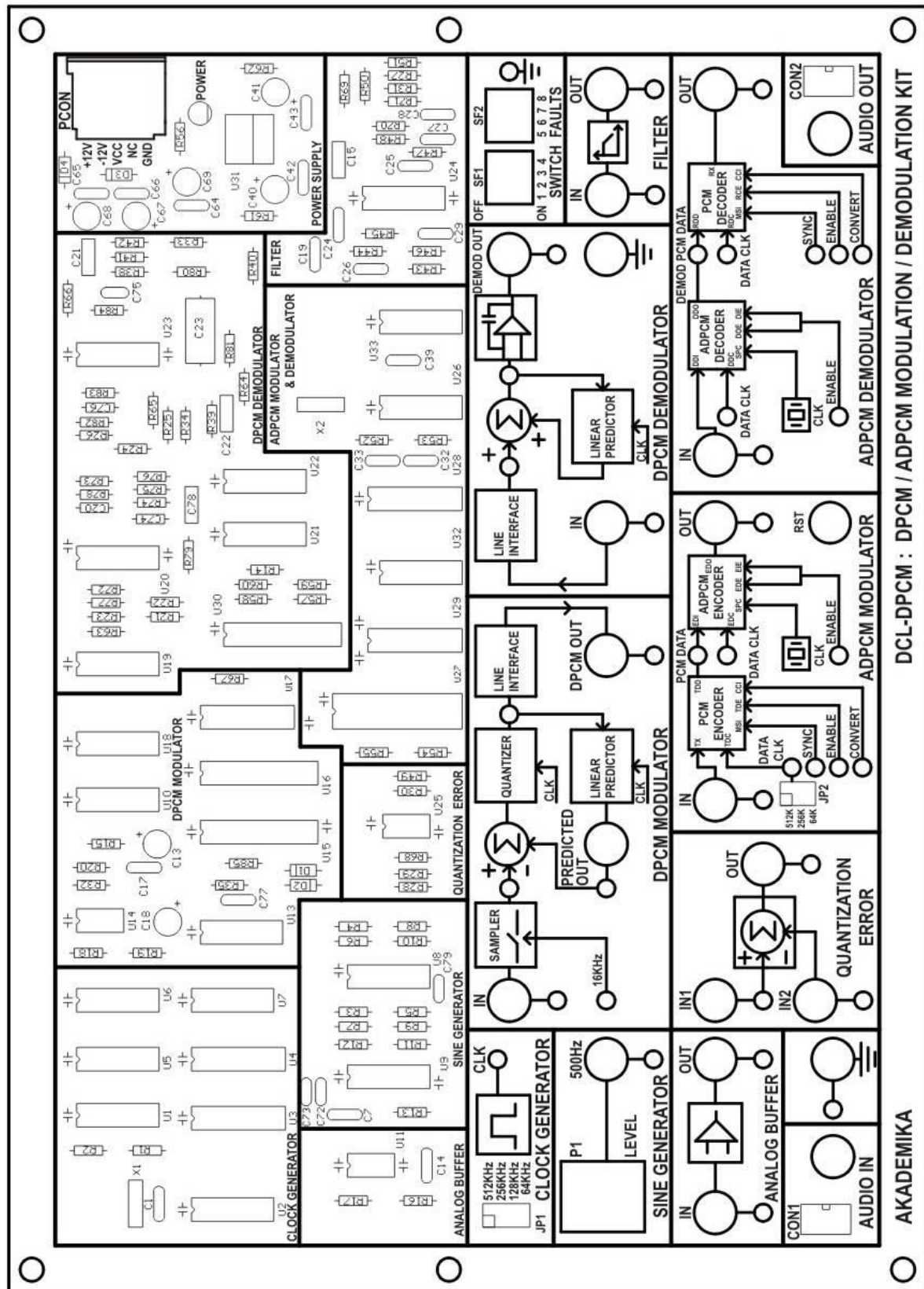
Accessories

DCL-DPCM			Quantity
1	Short Links PCS2 Patch cord Male To Male		10
2	ADCL-07 EXPT Manual		1
3	5 PIN DIN CONNECTOR Power Supply		1
4	Power cord		1

FEATURES

- Onboard signal generator, generating sine wave of frequency of 500Hz with variable amplitude
- Onboard selectable clock frequencies
- Sample and hold technique
- Single channel operation
- DPCM and ADPCM modulation and demodulation
- Quantization error
- Provision for interfacing with baseband modulation trainer kit for noise effect in DPCM and ADPCM (optional)
- Provision for interfacing Audio Input Output trainer kit for voice modulation (optional)
- Switch Faults: To study the different effects of switch faults, 8 switch faults are provided onboard of DCL-DPCM.
- Interconnection facilities: Sockets and connecting cords are provided for connections onboard and external connections.
- Test points: All relevant test points are provided onboard for observing intermediate signals. Observations are carried out on the oscilloscope.

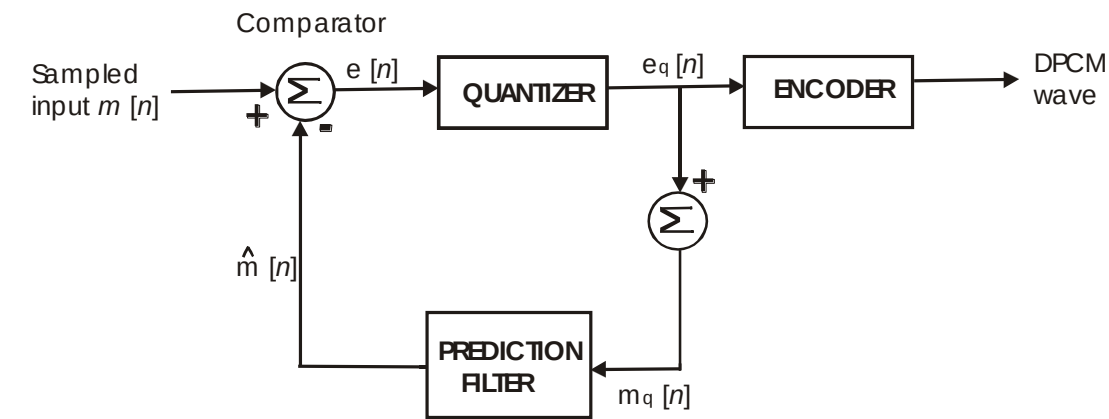
FUNCTIONAL BLOCK



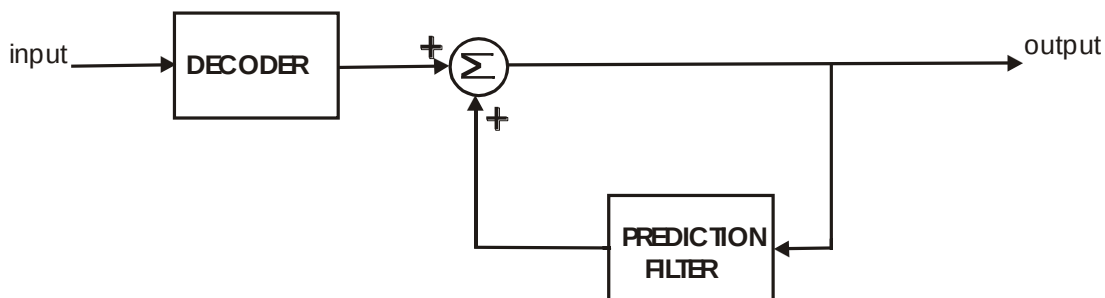
INTRODUCTION

AKADEMIKA's with the trainer, namely, "DCL-DPCM" initiates the user to study about the DPCM modulation and demodulation technique with quantization error. User can also study ADPCM modulation and demodulation using different CODECs.

DPCM And ADPCM Modulation And Demodulation Techniques Differential Pulse Code Modulation



(a)



(b)

DPCM system. (a) Transmitter. (b) Receiver.

At the time of PCM process, the differences between input sample signals are minimal. Differential PCM (DPCM) is designed to calculate this difference and then transmit this small difference signal instead of the entire input sample signal. Since the difference between input samples is less than an entire input sample, the number of bits required for transmission is reduced. This allows for a reduction in the throughput required to transmit voice signals.

When a voice or video signal is sampled at a rate slightly higher than the Nyquist rate as usually done in pulse code modulation, the resulting sampled signal is found to exhibit a high degree of correlation between adjacent samples. The meaning of this high correlation is that, in an average sense, the signal does not change rapidly from one sample to the

next, and as a result, the difference between adjacent samples has a variance that is smaller than the variance of the signal itself. When these highly correlated samples are encoded, as in the standard PCM system, the resulting encoded signal contains redundant information. This means that the symbols that are not absolutely essential to the transmission of information are generated as a result of the encoding process. By removing this redundancy before encoding, we obtain a more efficient coded signal, which is the basic idea behind differential pulse code modulation.

Now if we know the past behavior of a signal up to a certain point in time, we may use prediction to make an estimate of a future value of a signal. Suppose then a baseband signal $m(t)$ is sampled at the rate $f_s = 1/T_s$ to produce the sequence $\{m[n]\}$ whose samples are T_s seconds apart. The fact that it is possible to predict future values of the signal $m(t)$ provides motivation for the differential quantization scheme shown in the figure. In this scheme, the input signal to the quantizer is defined by

$$e[n] = m[n] - m'[n] \quad (i)$$

Which is the difference between the unquantized input sample $m[n]$ and a prediction of it, denoted by $m'[n]$. This predicted value is produced by using a linear prediction filter whose input, as we will see, consists of a quantized version of the input sample $m[n]$. The difference signal $e[n]$ is the predicted error, since it is the amount by which the prediction filter fails to predict the input exactly. By encoding the quantizer output, we obtain a variant of PCM known as differential pulse code modulation (DPCM).

The quantizer output may be expressed as

$$e_q[n] = e[n] + q[n] \quad (ii)$$

Where $q[n]$ is the quantization error. According to the figure, the quantizer output $e_q[n]$ is added to the predicted value $m'[n]$ to produce the prediction filter input

$$m_q[n] = m'[n] + e_q[n] \quad (iii)$$

Substituting equation (ii) into equation (iii), we get

$$m_q[n] = m'[n] + e[n] + q[n] \quad (iv)$$

However, from equation (i) we observe that the sum term $m'[n] + e[n]$ is equal to the input sample $m[n]$. Therefore we may simplify equation (iv) as

$$m_q[n] = m[n] + q[n] \quad (v)$$

Which represents a quantized version of the input sample $m[n]$. That is irrespective of the properties of the prediction filter, the quantized sample $m_q[n]$ at the prediction filter input differs from the original input sample $m[n]$ by the quantization error $q[n]$. Accordingly, if the prediction is good, the variance of the prediction error $e[n]$ will be smaller than the variance of $m[n]$, so that a quantizer with a given number of levels can be adjusted to produce a quantization error with a smaller variance than would be possible if the input sample $m[n]$ were quantized directly as in a standard PCM system.

The receiver for reconstructing the quantized version of the input is shown in the figure. It consists of a decoder to reconstruct the quantized error signal. The quantized version of the original input is reconstructed from the decoder output using the same prediction filter used in the transmitter. In the absence of channel noise, we find that the encoded signal at the receiver input is identical to the encoded signal at the transmitter output. Accordingly, the corresponding receiver output is equal to $m_q[n]$, which differs from the original input $m[n]$ only by the quantization error $q[n]$ incurred as a result of quantizing the prediction error $e[n]$.

From the forgoing analysis we observe that, in a noise free environment, the prediction filters in the transmitter and receiver operate in the same sequence of samples, $m_q[n]$. It is with this purpose in mind that a feedback path is added to the quantizer in the transmitter. Differential pulse code modulation includes delta modulation as a special case. In particular, comparing the DPCM system and the DM system, we find two important

differences, i.e. the use of one-bit (two level) quantizer in delta modulator and the replacement of the prediction filter by a single delay element (i.e., zero prediction order). Simply put, DM is the 1-bit version of DPCM. Note that unlike standard PCM system, the transmitters of both DPCM and DM involve the use of feedback.

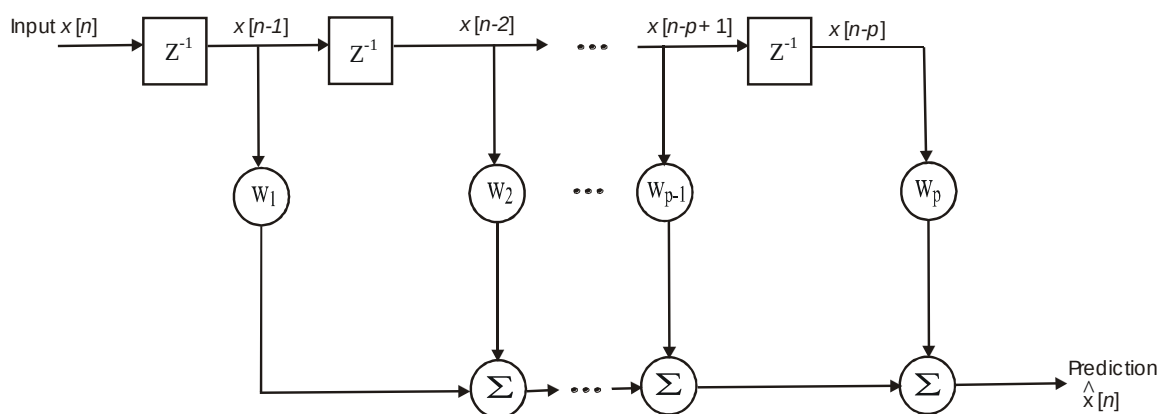
DPCM, like DM, is subject to slope overload distortion whenever the input signal changes too rapidly for the prediction filter to track it. Also like PCM, DPCM suffers from quantization noise.

Quantization Process

A continuous signal, such as voice, has a continuous range of amplitudes and therefore its samples have a continuous amplitude range. In other words, within the finite amplitude range of the signal, we find an infinite number of amplitude levels. It is not necessary in fact to transmit the exact amplitudes of the samples. Any human sense (the ear or the eye), as ultimate receiver, can detect only finite intensity differences. This means that the original continuous signal may be approximated by a signal constructed of discrete amplitudes selected on a minimum error basis from an available set. The existence of a finite number of discrete amplitude levels is a basic condition of pulse code modulation. Clearly, if we assign the discrete amplitude levels with sufficiently close spacing, we make the approximated signal practically indistinguishable from the original continuous signal.

Quantizers can be of uniform and nonuniform type. In a uniform quantizer, the representation levels are uniformly spaced; otherwise, the quantizer is nonuniform.

Linear Prediction



Block diagram of a linear prediction filter of order p .

Consider a finite-duration impulse response (FIR) discrete time filter configured as in the figure, which involves the use of three functional blocks:

- Set of p unit-delay elements, each of which is represented by z^{-1} .
- Set of multipliers involving the filter coefficients $w_1, w_2, \dots, w_p$.
- Set of “adders” used to sum the scaled versions of the delayed inputs $x[n-1], x[n-2], \dots, x[n-p]$ to produce the output $\hat{x}[n]$, or more precisely, the linear prediction of the input, is thus defined by the convolution sum

$$\hat{x}[n] = \sum_{k=1}^p w_k x[n-k] \quad (i)$$

Where p , the number of unit-delay elements, is called the prediction order

The actual sample at time nT_s is $x[n]$. The prediction error, denoted by $e[n]$, is defined as the difference between $x[n]$ and the prediction $\hat{x}[n]$, as shown by

$$e[n] = x[n] - \hat{x}[n] \quad (\text{ii})$$

The design objective is to choose the filter coefficients w_1, w_2, w_p so as to minimize an index of performance, J , defined as the mean square error:

$$J = E[e^2[n]] \quad (\text{iii})$$

Substituting equations (i) and (ii) into (iii) and then expanding terms, we may reformulate the index of performance as

$$J = E[x^2[n]] - 2 \sum_{k=1}^p w_k E[x[n] x[n-k]] + \sum_{j=1}^p \sum_{k=1}^p w_j w_k E[x[n-j] x[n-k]] \quad (\text{iv})$$

We assume that the input signal $x(t)$ is the sample function of the stationary process $X(t)$ of zero mean; that is, $E[x[n]]$ is zero for all n . Define

$$\begin{aligned} \sigma^2_x &= \text{variance of a sample of the process } X(t) \text{ at time } nT_s \\ &= E[x^2[n]] - (E[x[n]])^2 \\ &= (E[x[n]])^2 \\ R_x(kT_s) &= \text{autocorrelation of the process } X(t) \text{ for a lag of } kT_s \\ &= R_x[k] \\ &= E[x[n] x[n-k]] \end{aligned}$$

Accordingly, we may rewrite equation (iv) in the simplified form

$$J = \sigma^2_x - 2 \sum_{k=1}^p w_k R_x[k] + \sum_{j=1}^p \sum_{k=1}^p w_j w_k R_x[k-1] \quad (\text{v})$$

Hence differentiating the index of performance J with respect to the filter coefficient w_k , setting the result equal to zero, and then rearranging terms, we obtain

$$\sum_{j=1}^p w_j R_x[k-1] = R_x[k] = R_x[-k], \quad k = 1, 2, \dots, p \quad (\text{vi})$$

The optimality equations (vi) are called the Wiener-Hopf equations for linear prediction. We find it convenient to reformulate the Wiener-Hopf equations (vi) in matrix form. Let

$$\begin{aligned} w_o &= p\text{-by-1 optimum coefficient vector} \\ &= [w_1, w_2, \dots, w_p]^T \\ r_x &= p\text{-by-1 autocorrelation vector} \\ &= [R_x[1], R_x[2], \dots, R_x[p]]^T \\ R_x &= p\text{-by-1 autocorrelation matrix} \end{aligned}$$

$$= \begin{bmatrix} R_x[0] & R_x[1] & \dots & R_x[p-1] \\ R_x[1] & R_x[0] & \dots & R_x[p-2] \\ \vdots & \vdots & \ddots & \vdots \\ R_x[p-1] & R_x[p-2] & \dots & R_x[0] \end{bmatrix}$$

We may thus simplify the set of equations (vi) as

$$R_x w_0 = r_x \quad (\text{vii})$$

We assume that the autocorrelation matrix R_x is nonsingular, so that its inverse exists. We may then solve equation (vii) for the coefficient vector by w_0 multiplying both sides of this equation by the inverse matrix R_x^{-1} , obtaining the optimum solution

$$w_0 = R_x^{-1} r_x \quad (\text{viii})$$

Note that all the elements on the main diagonal of the autocorrelation matrix R_x are equal to $R_x[0] = \sigma^2 x$, and the elements on any other diagonal parallel to the main diagonal are also equal. A square matrix having this property is said to be Toeplitz, which is a direct consequence of the assumption that the input signal $x(t)$ is drawn from a stationary process. The practical significance of the Toeplitz property is that the correlation matrix R_x is uniquely defined by the set of autocorrelation value $R_x[0], R_x[1], \dots, R_x[p-1]$. The autocorrelation vector r_x is defined by the set of autocorrelation value $R_x[1], R_x[2], \dots, R_x[p]$. It follows therefore that the p filter coefficients of the linear optimum predictor are uniquely defined by the variance $\sigma^2 x = R_x[0]$ and p values of the autocorrelation function of the process $X(t)$ for lags of T_s

$$2T_s, pT_s.$$

The minimum mean-square value of the prediction error is obtained by substituting Equation (vi) into (v), which yields (after simplification)

$$J_{\min} = \sigma^2 x - r_x^T R_x^{-1} r_x \quad (\text{ix})$$

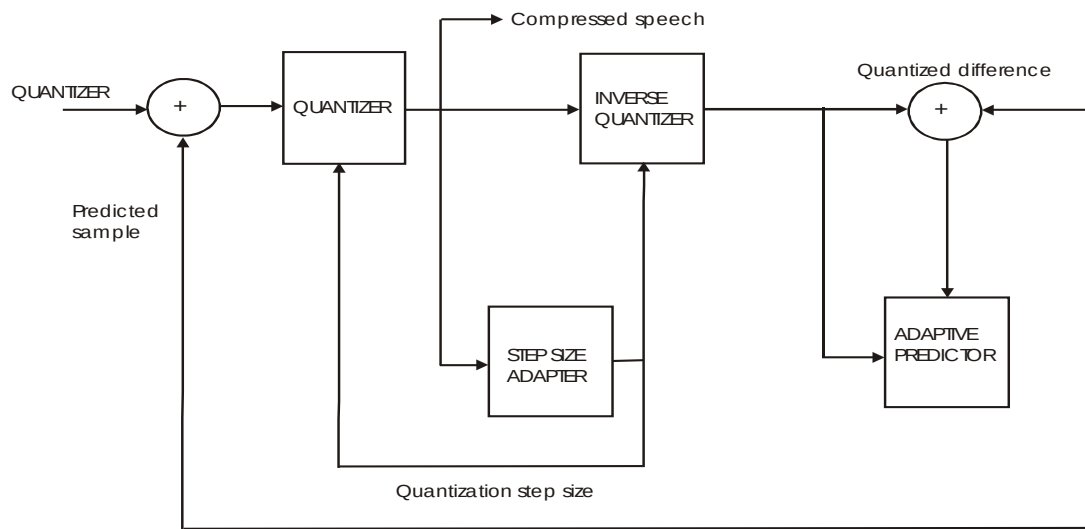
The quadratic $r_x^T R_x^{-1} r_x$ term is always positive. Accordingly, the mean- Square error $J_{\min}$ of the optimum linear predictor defined by equation (ix) is Always less than the variance $\sigma^2 x$ of the input sample that is being Predicted

Adaptive Differential Pulse Code Modulation

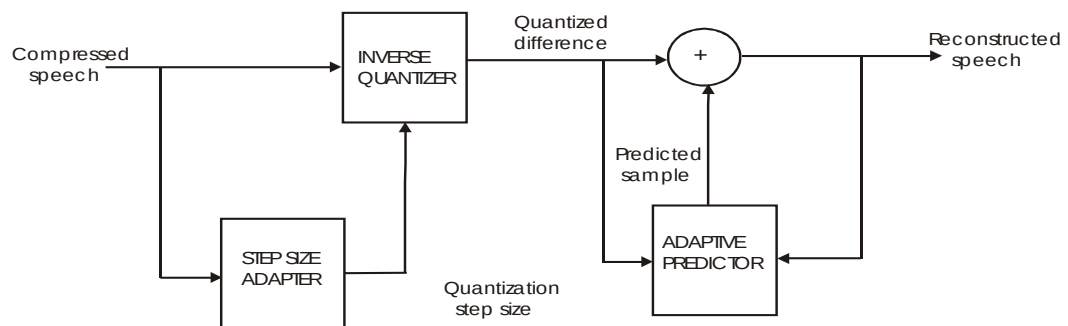
The use of PCM for pulse coding at the standard rate of 64kb/s demands a high channel bandwidth for its transmission. In certain applications, however, such as secure transmission over radio channels that are inherently of low capacity, channel bandwidth is at a premium. In applications of this kind, there is a definite need of speech coding at low bit rates, while maintaining acceptable fidelity or quality of reproduction.

For coding speech at low bit rates, a waveform coder of prescribed configuration is optimized by exploiting both statistical characterization of speech waveforms and properties of hearing. In particular, the design philosophy has two aims in mind:

1. To remove redundancies from the speech signal as far as possible.
2. To assign the available bits to code the non-redundant parts of speech signal in a perceptually efficient manner.



ADPCM-ENCODER



ADPCM-DECODER

As we strive to reduce the bit rate from 64kb/s (used in standard) PCM to 32, 16, 8 and 4kb/s, the schemes used for redundancy removal and bit assignment become increasingly more sophisticated. As a rule of thumb, in 64 to 8kb/s range, the computational complexity (measured in terms of multiply – add operations) required to code speech increases by an order of magnitude when the bit rate is halved, for approximately equal speech quality.

Adaptive Differential Pulse Code modulation permits the coding of speech at lower bit rates through the combined use of adaptive quantization and adaptive prediction. The term adaptive used herein means being responsive to changing level and spectrum of the input speech signal. The variance of performance with speakers and speech material, together with variations in signal level inherent in the speech communication process, make the combined use of adaptive quantization and adaptive prediction necessary to achieve best performance over a wide range of speakers and speaking situations.

Adaptive quantization refers to a quantizer that operates with a time varying step-size $\Delta[n]$. At any given sampling instant identified by the index n , the adaptive quantizer is assumed to have a uniform transfer characteristic. The step size $\Delta[n]$ is varied so as to match the variance σ_M^2 of the input sample $m[n]$. In particular, we write

$$\Delta[n] = \sqrt{\sigma_M[n]}$$

Where, $\sqrt{}$ is constant, and $\sigma_M[n]$ is an estimate of the standard deviation $\sigma_M[n]$ (i.e., square root of the variance σ_M^2).

Linear Adaptive Prediction

The use of equation (viii) in the linear predictor, for calculating the weight vector of a linear predictor requires knowledge of the autocorrelation function $R_x[k]$ of the input sequence $\{x[n]\}$ for lags $k = 0, 1, \dots, p$, where p is the prediction order. What if knowledge of $R_x[k]$ for varying k is not available? In these situations, which occur frequently in practice, we may resort to the use of an adaptive predictor.

The predictor is adaptive in the following sense:

1. Computation of the tap weights w_k , $k=1, 2, \dots, p$, proceeds in a “recursive” manner, starting from some arbitrary initial value of the tap weights.
2. The algorithm used to adjust the tap weights (from one iteration to the next) is “self designed,” operating solely on the basis of available data.

The aim of the algorithm is to find the minimum point of the bowl-shaped error surface that describes the dependence of the cost function j on the tap weights. It is therefore intuitively reasonable that successive adjustments to the tap weights of the predictor be made in the direction of the steepest descent of the error surface, that is, in a direction opposite to the gradient vector whose elements are defined by

$$g_k = \partial j / \partial w_k, \quad k = 1, 2, \dots, p \quad (i)$$

This is indeed the idea behind the method of steepest descent. Let $w_k[n]$ denote the value of the k th tap-weight at iteration n . Then the updated value of this weight at iteration $n + 1$ is defined by

$$w_k[n + 1] = w_k[n] - 1/2 \mu g_k, \quad k = 1, 2, \dots, p \quad (ii)$$

Where μ is a step-size parameter that controls the speed of adaptation, and the factor $1/2$ is included for convenience of presentation. Differentiating the cost function j of equation (v) of linear predictor, with respect to w_k , we readily find that

$$\begin{aligned} g_k &= -2 R_x[k] + 2 \sum_{j=1}^p w_j R_x[k-j] \\ &= -2E[x[n] x[n-k]] + 2 \sum_{j=1}^p w_j E[x[n-j] x[n-k]], \quad k = 1, 2, \dots, p \end{aligned} \quad (iii)$$

This formula for g_k could do with further simplification, which is achieved by using instantaneous values as estimates of the autocorrelation functions $R_x[k]$ and $R_x[k-j]$. That is, we ignore the expectation operators in equation (iii) to facilitate the adaptive process on a step-by-step basis. We may thus express the corresponding estimate of g_k at iteration n as

$$g'_k = -2x[n]x[n-k] + 2 \sum_{j=1}^p w_j[n]x[n-j]x[n-k], \quad k = 1, 2, \dots, p \quad (iv)$$

Note that for an input $x[n]$ drawn from a stationary process the gradient g_k is a deterministic quantity, whereas the estimate $g'_k[n]$ is the sample value of a random variable.

In any event, substituting equation (iv) into (ii) and factoring the common term $x[n-k]$, we may write

$$w'_k[n+1] = w'_k[n] + \mu x[n-k] (x[n] - \sum_{j=1}^p w'_j[n] x[n-j])$$

$$= w'_k[n] + \mu x[n-k] e[n], \quad k = 1, 2, \dots, p \quad (v)$$

Where $e[n]$ is the prediction error defined as

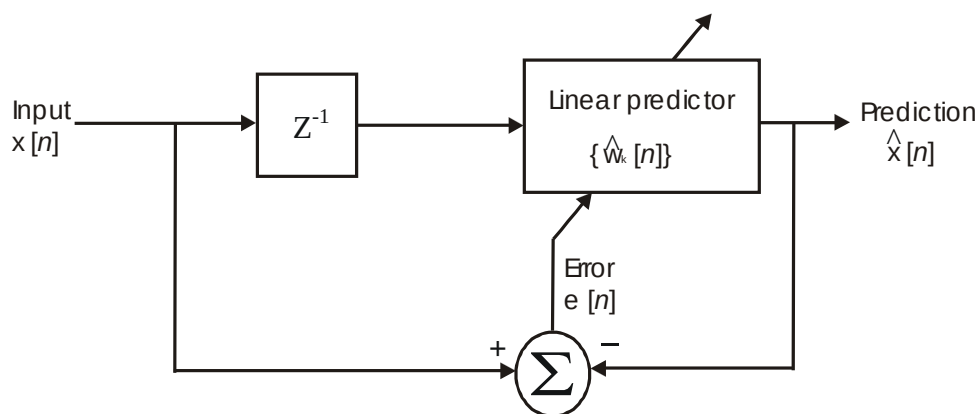
$$e[n] = x[n] - \sum_{j=1}^p w'_j[n] x[n-j] \quad (vi)$$

In equations (v) and (vi), we have used w'_k as an estimate of the k th tap-weight to distinguish it from the actual value w_k . Note also that $x[n]$ plays the role of a “desired response” for computing the recursive adjustments applied to the tap-weights of the predictor.

Equations (v) and (vi) constitute the popular least-mean-square (LMS) algorithm for linear adaptive prediction, the operation of which is depicted in the given figure. The reason for popularity of this adaptive filtering algorithm is the simplicity of its implementation. In particular, the computational complexity of the algorithm, measured in terms of the number of additions and multiplications, is linear in the prediction order p .

The LMS algorithm is a stochastic adaptive filtering algorithm, stochastic in the sense that, starting from the initial condition defined by $\{w_k[0]\}_{p_j=1}$, it seeks to find the minimum point of the error surface by following a zigzag path. Moreover, it never finds this minimum point exactly. Rather, it executes a random motion

Around the minimum point of the error surface, once steady-state conditions are established



Block diagram illustrating the linear prediction process

CIRCUIT DESCRIPTION

DPCM MODULATOR

- **Clock Generator**

The crystal oscillator generates a 2.048MHz clock. 1.024MHz clock is generated by dividing the 2.048MHz clock. 512 KHz, 256 KHz, 128 KHz and 64 KHz clocks are generated using U (74HCT393) a 4-bit binary counter.

- **Timing Logic**

This logic generates the various timing signals for the transmitter. The operation of ADC673 is controlled by two inputs: CONVERT and /DATA ENABLE. The transmitted clock is fed to a 4-bit binary counter (IC74LS393) whose output is AND'ed using (IC74HC08). A pulse (plrst) is generated which is further fed to D flip-flop to generate the CONVERT pulse and the /DATA ENABLE signal.

- **Sine Wave Generator**

8 KHz frequency is fed to serial to parallel shift register (IC74HC164) which generates the sine wave, by serial shift operations. The serial to parallel shift register (IC74HC164) has resistive ladder network at the output. For 8 shifts of register, one staircase sine wave is generated. So if 8KHz clock is fed to the shift register, 500Hz sine wave is generated with amplitude variable from 0 to 4Vpp.

- **Sample And Hold Logic**

This logic samples the information signal with the clock frequency of 16KHz. (IC4016) is used for sampling. It is a quad bilateral switch intended for the transmission or multiplexing of analog or digital signals.

- **Analog To Digital Converter**

The analog to digital converter converts the analog samples to digital bits. This device performs both the quantizing and encoding operations. The analog to digital converter IC, AD673 forms the heart of this logic. The timing of the A/D conversion is controlled by the convert pulse, which is generated in the timing logics. As soon as the data is ready, the AD673 gives a data ready (DR active low) output. The data latch is used for latching the valid outputs of the A/D converter. The timing and latching interval of the data latch is controlled by the latch enable signal (74HCT374). Each sample is coded to a 8 bit data by AD673.

These coded data is fed to a parallel to serial shift register, which gives a serial 8-bit quantized data.

- **Quantizing**

. The signal level of the input signal assumed to vary from 0 volt to 4.96 volts. The entire level is dividing to 128 uniform steps. Each step corresponds to

$$4.96\text{V}/128 = 40\text{mv}$$

The quantizing level is chosen to be the midway of the steps. If the signal level falls below the quantizing level, then the signal is rounded off to the lower level. If the signal level falls above the quantizing level, and then the signal level is rounded off to the upper level. This type of quantizing is called " uniform quantizing ", where the step levels are uniform. Then corresponding to the level chosen, code words are assigned to the samples. The code words vary from 0000000 to 1111111 for the 0 to 4.96V. Thus the analog to digital converter assigns the code words for all the samples. For further understanding of Analog to Digital conversion principles, refer the datasheet of AD673.

- **Linear Predictor**

D flip-flops (74HCT74) are used as delay elements to give two bit delays in the quantized data. The two delayed data's are ORed using (IC74HC32). The ORed data is fed to the integrator circuit, which is built around (ICTL084).

- **Comparator**

The comparator circuit is built around (ICLM311). This circuit compares the two inputs and the differenced output is fed to the quantizer.

DPCM DEMODULATOR

- **Linear Predictor**

Refer to the circuit diagram as shown later in this manual sheet 5 of 8 of ADCL-07. D flip-flops (74HCT74) are used as delay elements to give two bit delays. The received data and the delayed is ORed using (74HC32). The delay element is placed in feedback loop with the OR circuit.

- **Integrator**

The integrator is built around (ICTL084). Integrator output is the integration of the input signal applied.

- **Butterworth Filter**

The butter-worth low pass filter is built around (ICTL084), which filters filter outs the sampling frequency components from the demodulation output.

- **ADPCM Modulator**

Refer to the circuit diagram as shown later in this manual sheet 7 of 8 of ADCL 07. CODEC (MC145502) is used for analog to digital conversion. The whole process takes place inside the CODEC. Present techniques of voice communication standards such as A-law / μ -law companded PCM voice coding at 64kbps. If the amplitude of the signal is small; quantization levels have to be closely spaced. This gives proper resolution. But if the signal amplitude is large than this fine resolution will result in increasing the number of code bits. The analog input is fed into the CODEC; the output is in digital form. The digital data output is in Pulse Code Modulated form. The CODEC chips used exhibit both A-law and μ -law companding techniques. Here, we select this A-law and μ -law companding with the help of a switch.

The PCM data is fed to the ADPCM CODEC (MC145532). This CODEC is used to reduce the data rate required to transmit a PCM encoded analog / voice signal while maintaining the voice fidelity and intelligibility of the PCM signal. It also uses a filter to attempt to predict the next PCM input value based on previous PCM input is values. The error between the predicted and the true PCM input value is the information that is sent to the other end of the line. The adaptive filter adapts to the statistics of the signals presented to it. Here also the whole process takes place inside the CODEC.

- **ADPCM Demodulator**

CODEC (MC145532) is used for ADPCM demodulation. The ADPCM data is fed into this CODEC along with clocks for synchronization. Demodulated PCM data is derived. This PCM data is fed into the PCM CODEC filter (MC145502). The received digital data is again converted into analog form by the chip.

The clock considerations and other features of these CODEC chips used in this kit can be studied in details from the data sheets provided with the manual.

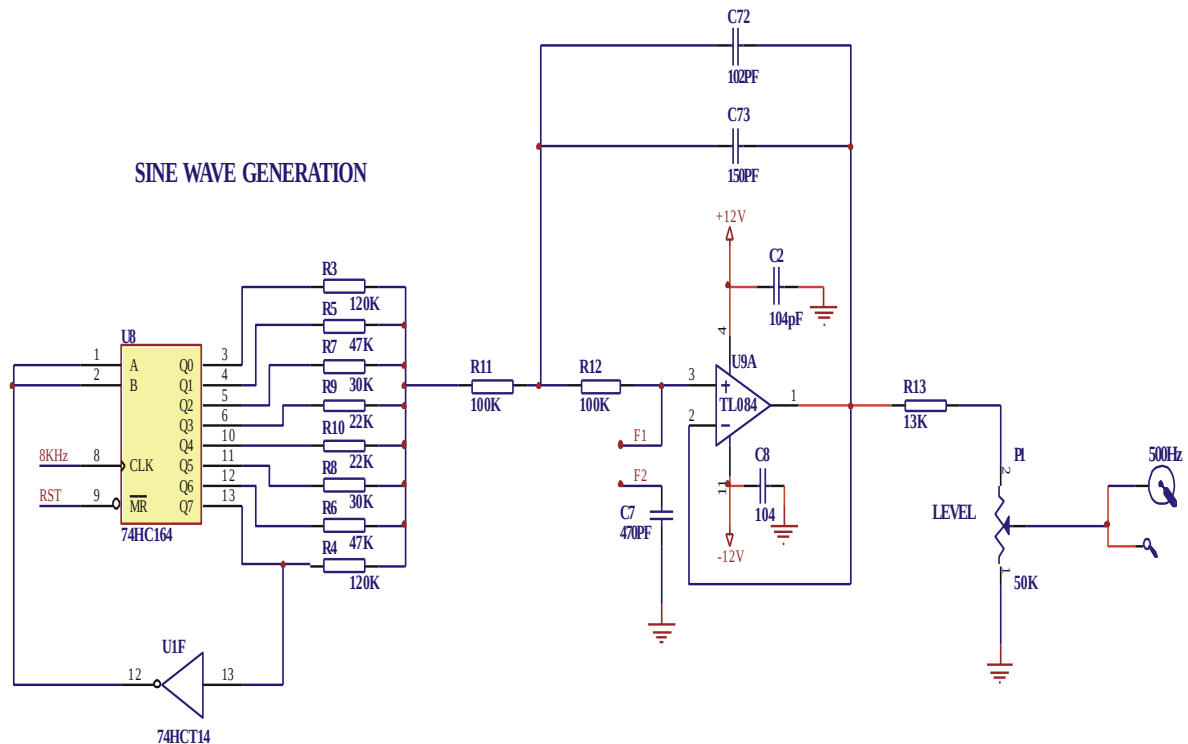
- **Audio In And Audio Out**

These connectors are provided to connect Audio input and Audio output interfacing cards to DCL-DPCM for voice communication.

SWITCH FAULTS

Switch fault section on DCL-DPCM provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the DCL-DPCM is fully operational, all switches should be set to OFF position before use.

1) Effect Of Switch Fault 1(DCL-DPCM) In Sine Wave Generation Section



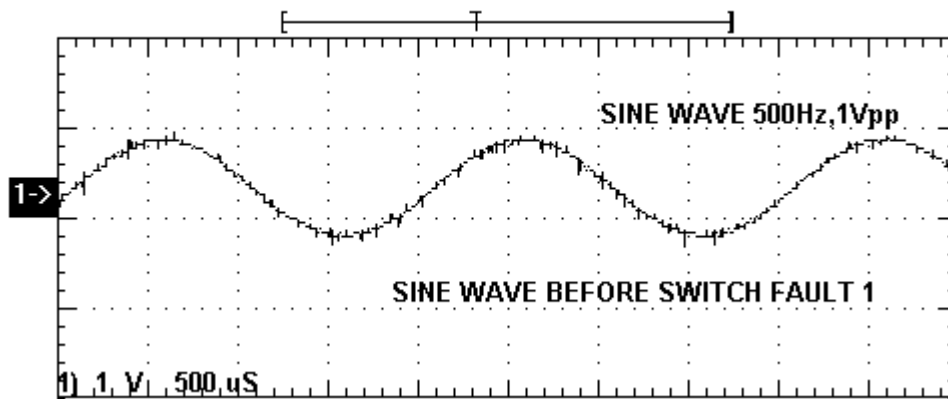
DESCRIPTION

This circuit generates 500Hz sine wave. 8KHz frequency is fed to serial to parallel shift register (IC74HC164) which generates the sine wave, by serial shift operations. The serial to parallel shift register (IC74HC164) has resistive ladder network at the output. For 8 shifts of register, one staircase sine wave is generated. So if 8KHz clock is fed to the shift register, 500Hz sine wave is generated with amplitude variable from 0 to 4Vpp.

PROCEDURE

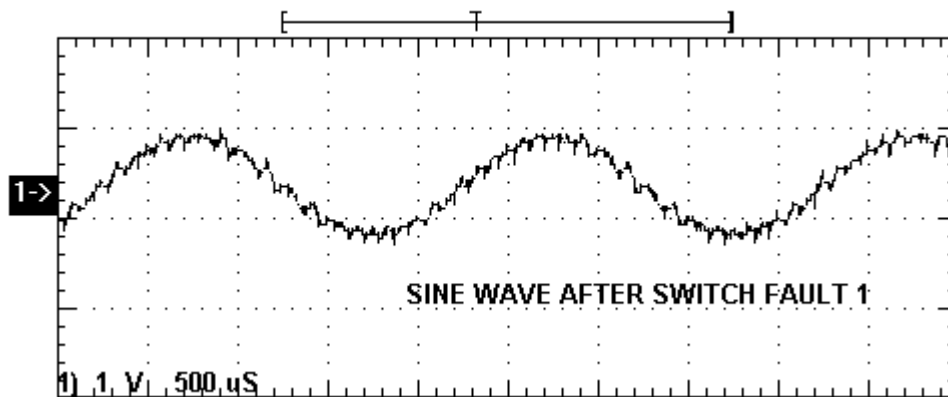
1. First keep switch **1 (SF1)** to OFF position. Observe the waveform at **500Hz** output post of the SINE GENERATOR.

Waveform before Switch Fault



Now put the switch **1 (SF1)** in the Switch Fault to ON position and observe its effect.

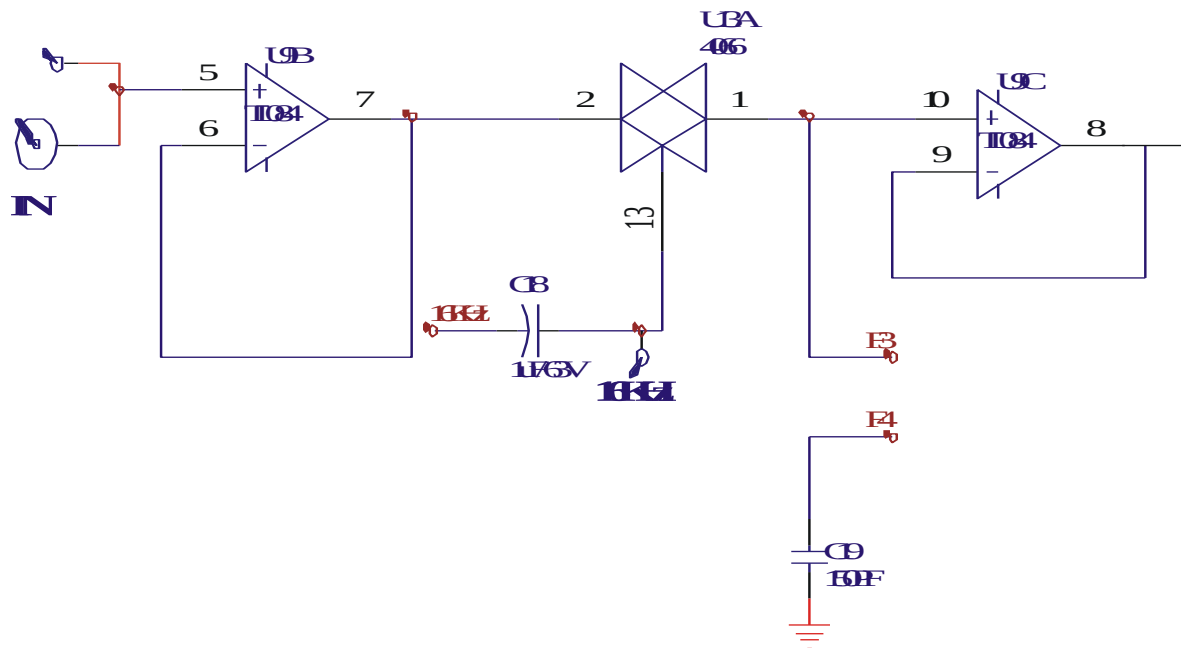
Waveform after Switch Fault



Action And Effect

When we put the **1** of **SF1** (DCL-DPCM) in Switch Fault section to ON position, it opens the filtering capacitor C7 of U9 of the sine wave generator circuit due to which the sine wave gets distorted.

2) Effect of Switch Fault 2 (DCL-DPCM) In Sampler Section



DESCRIPTION

This circuit is used for sampling, this logic samples the information signal with the clock frequency of 16KHz. IC 4016 is used for sampling. It is a quad bilateral switch intended for the transmission or multiplexing of analog or digital signals.

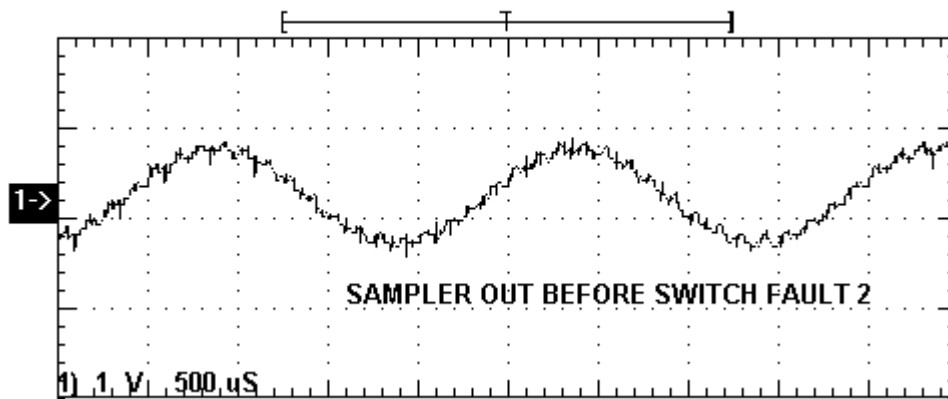
PROCEDURE

First keep switch **2 (SF1)** to OFF position. Observe the waveform at the given output test point of the sampler in the DPCM MODULATOR section.

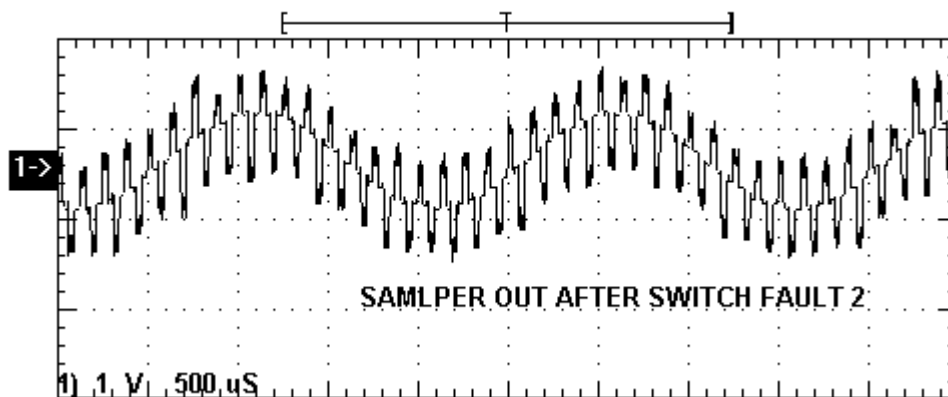
1. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
2. Keep the clock frequency at 512 KHz, by changing the jumper position in the clock generator section.
3. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.

DPCM modulation:

4. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect the **OUT** post of buffer to the **IN** DPCM modulator section.
6. Observe the sample output at the given test point. The input signal is sampled at the clock frequency of 16 KHz.

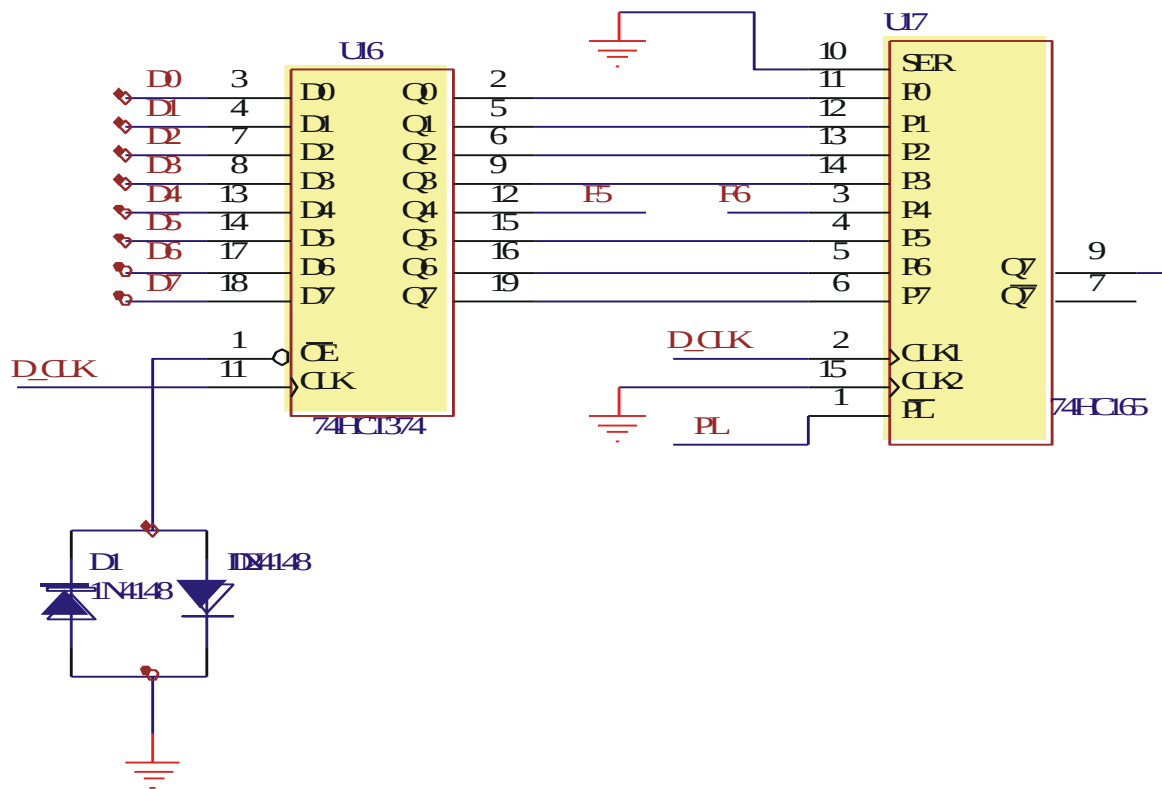
Waveform before Switch Fault:

Now put the switch 2 (**SF1**) in the Switch Fault to ON position and observe its effect.

Waveform after Switch Fault**Action And Effect**

When we put the 2 of **SF1** (DCL-DPCM) in Switch Fault section to ON position, the capacitor C19 is opened due to which the sampled signal is distorted

3) Effect of Switch Fault 3 (DCL-DPCM) In Quantization Section



DESCRIPTION

The data latch (74HCT374) is used for latching the valid outputs (i.e. the quantized data) of the A/D converter. The timing and latching interval of the data latch is controlled by the latch enable signal. Each sample is coded to a 8 bit data by AD673. These coded data is fed to a parallel to serial shift register (74HC165), which gives a serial 8-bit quantized data.

PROCEDURE

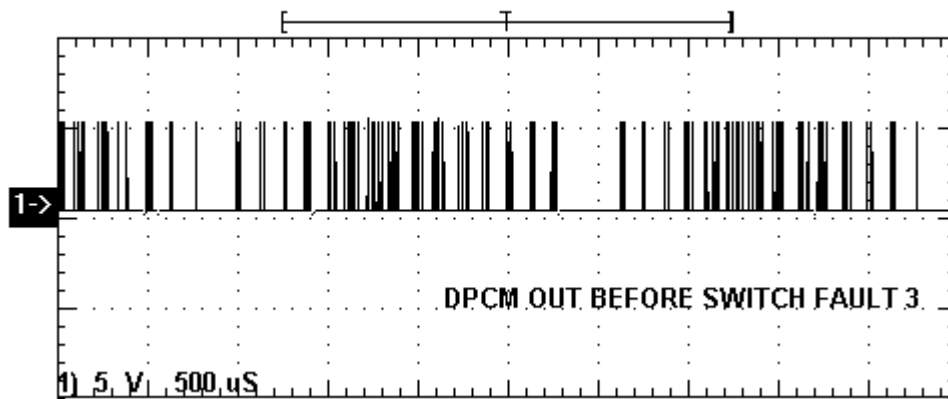
First keep switch 3 (**SF1**) to OFF position. Observe the waveform at **DPCM** outpost of the DPCM modulator section.

1. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
2. Keep the clock frequency at 512 KHz, by changing the jumper position in the clock generator section.
3. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.

DPCM modulation:

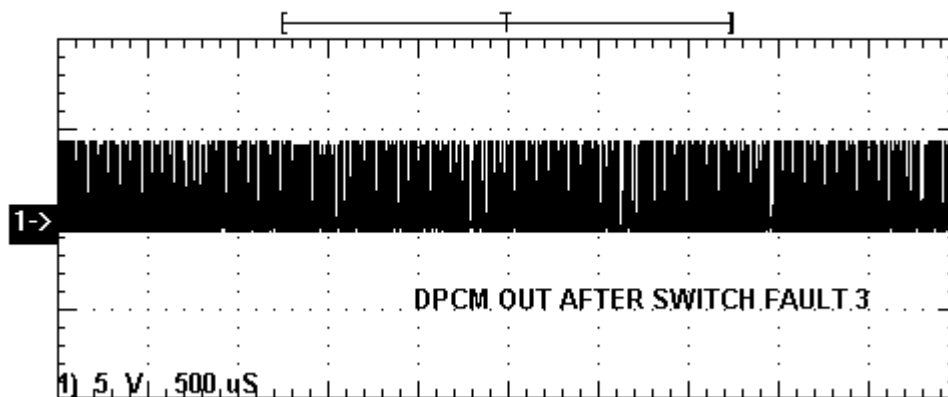
4. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect the **OUT** post of buffer to the **IN** DPCM modulator section.
6. Observe the differential pulse code modulated data (DPCM) at the **OUT** post of the DPCM modulator section.

Waveform before Switch Fault



Now put the switch **3 (SF1)** in the Switch Fault to ON position and observe its effect.

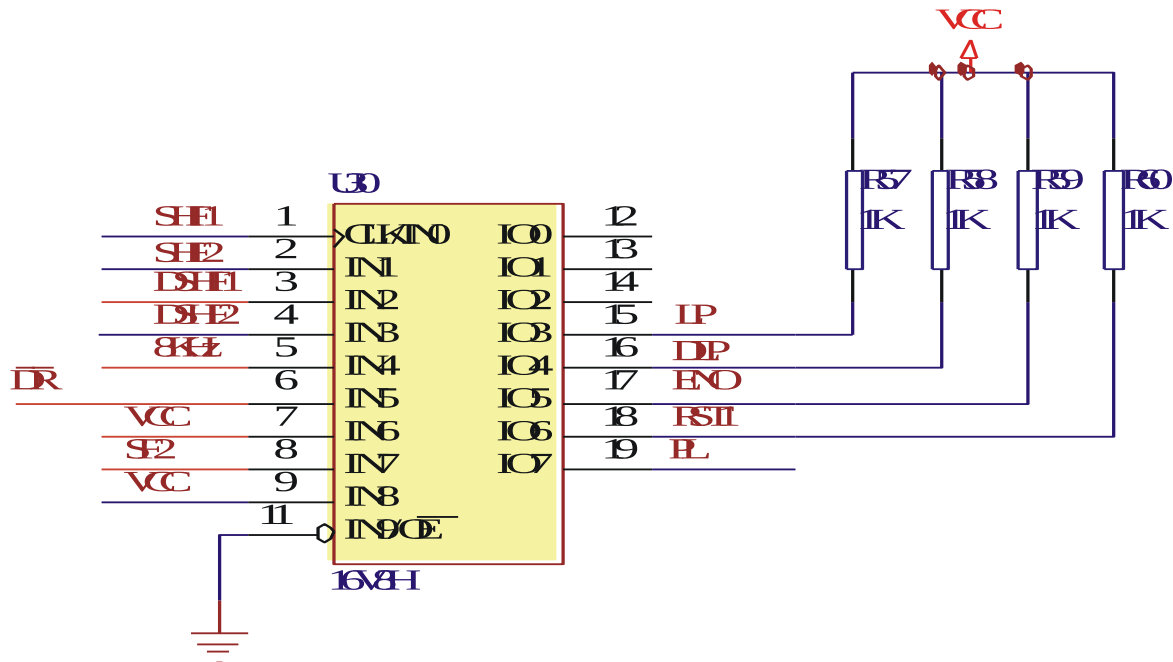
Waveform after Switch Fault



Action And Effect

When we put the **3** of **SF1** (DCL-DPCM) in Switch Fault section to ON position, it opens pin no.12 of U16 (74HCT374) and pin no. 3 of U17 (74HC165). Due to which there is loss in data. The DPCM output gets disturbed.

4) Effect of Switch Fault 4 (DCL-DPCM) In Data Generation Section



DESCRIPTION

The Digital logic is performed using the PAL IC.

PROCEDURE:

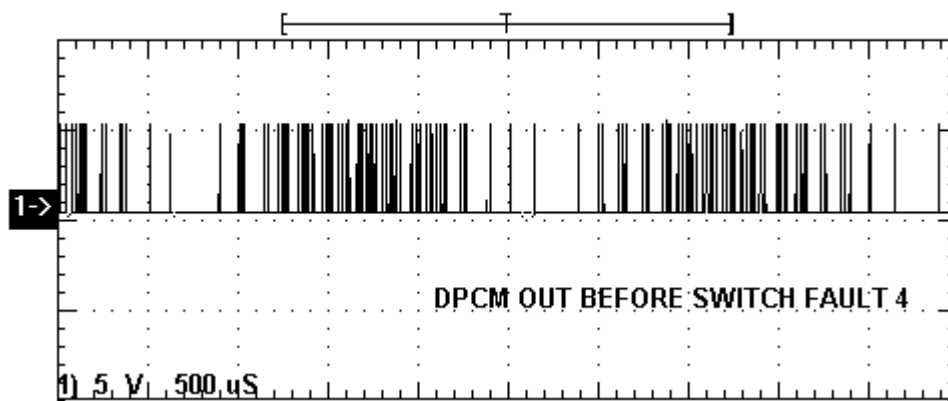
First keep switch 4 (**SF1**) to OFF position. Observe the waveform at **DPCM** outpost of the DPCM modulator.

1. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
2. Keep the clock frequency at 512 KHz, by changing the jumper position in the clock generator section.
3. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.

DPCM modulation

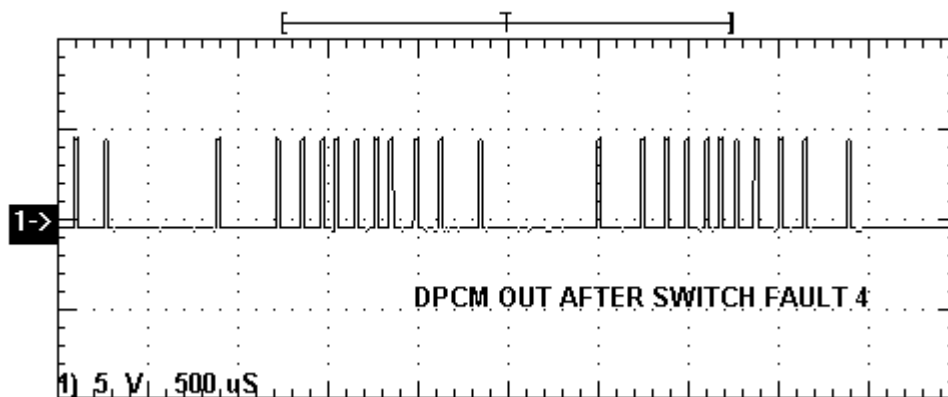
4. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect the **OUT** post of buffer to the **IN** DPCM modulator section.
6. Observe the differential pulse code modulated data (DPCM) at the **OUT** post of the DPCM modulator section.

Waveform before Switch Fault



Now put the switch **4 (SF1)** in the Switch Fault to ON position and observe its effect.

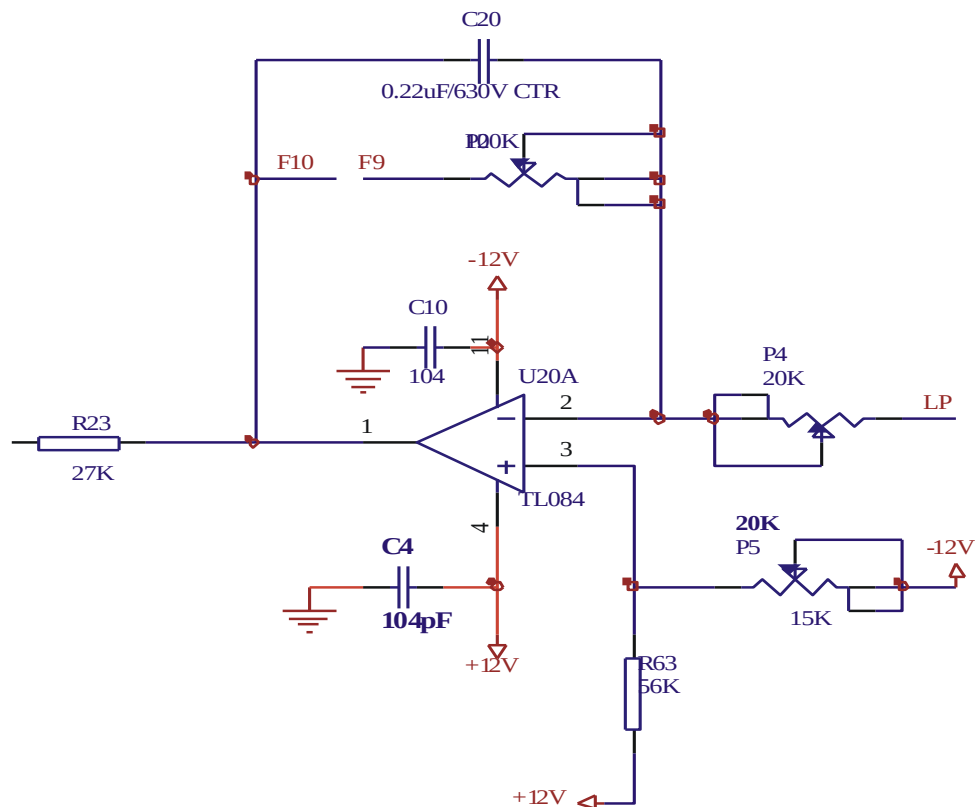
Waveform after Switch Fault



Action and Effect

The Digital logic is performed using the PAL IC. When we put the **4** of **SF1** (DCL-DPCM) in Switch Fault section to ON position, the PLRST signal going to pin no.8 of U30 (PAL16V8) is opened due to which the enable signal (PL) required for parallel to serial conversion is not obtained. This gives us a disturbed DPCM data.

5) Effect of Switch Fault 5 (DCL-DPCM) In Data Generation Section



DESCRIPTION

The predicted data is integrated through the integrator circuit.

PROCEDURE

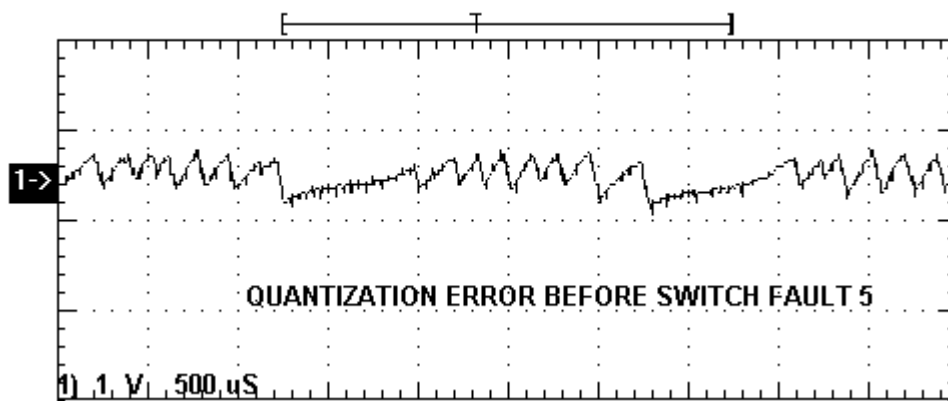
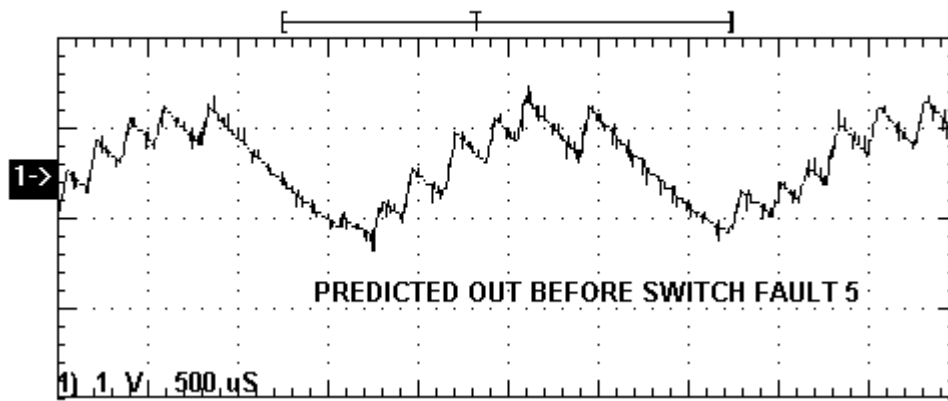
First keep switch **5 (SF2)** to OFF position. Observe the waveform at **PREDICTED OUT** outpost of the DPCM modulator.

1. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
2. Keep the clock frequency at 512 KHz, by changing the jumper position in the clock generator section.
3. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.

DPCM modulation:

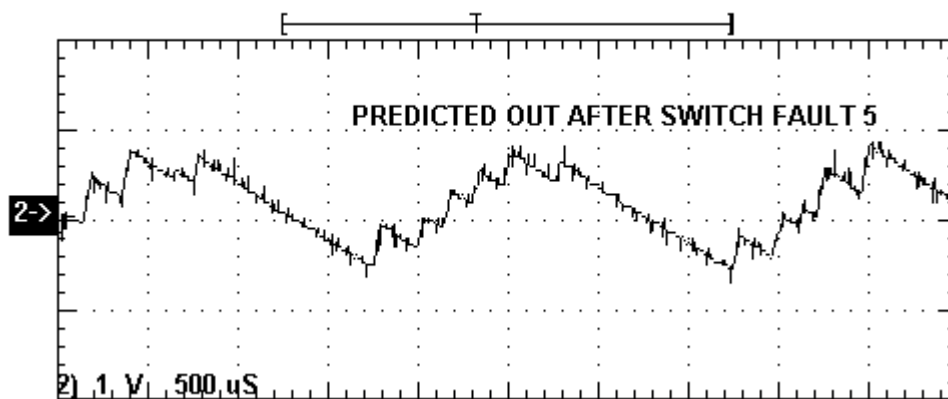
4. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect the **OUT** post of buffer to the **IN** DPCM modulator section.
6. Observe the sample output at the given test point. The input signal is sampled at the clock frequency of 16 KHz.
7. Observe the linear predictor output at the **PREDICTED OUT** post of the linear predictor in the DPCM modulator section.

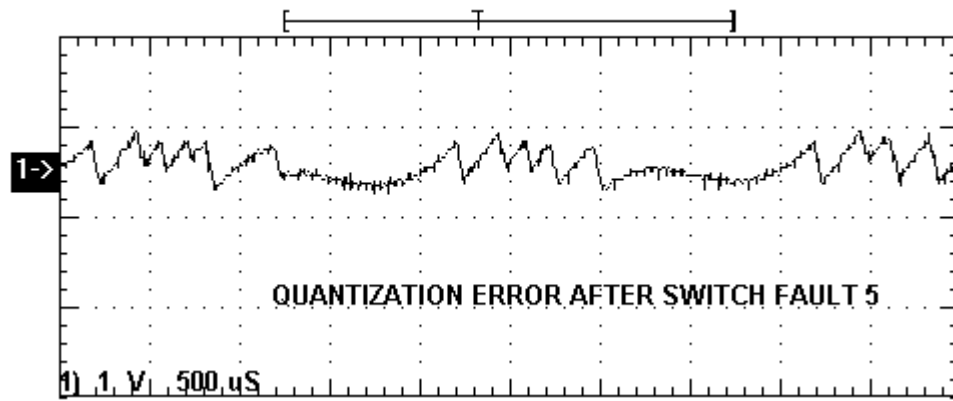
Waveform before Switch Fault



Now put the switch 5 (SF2) in the Switch Fault to ON position and observe its effect.

Waveform after Switch Fault

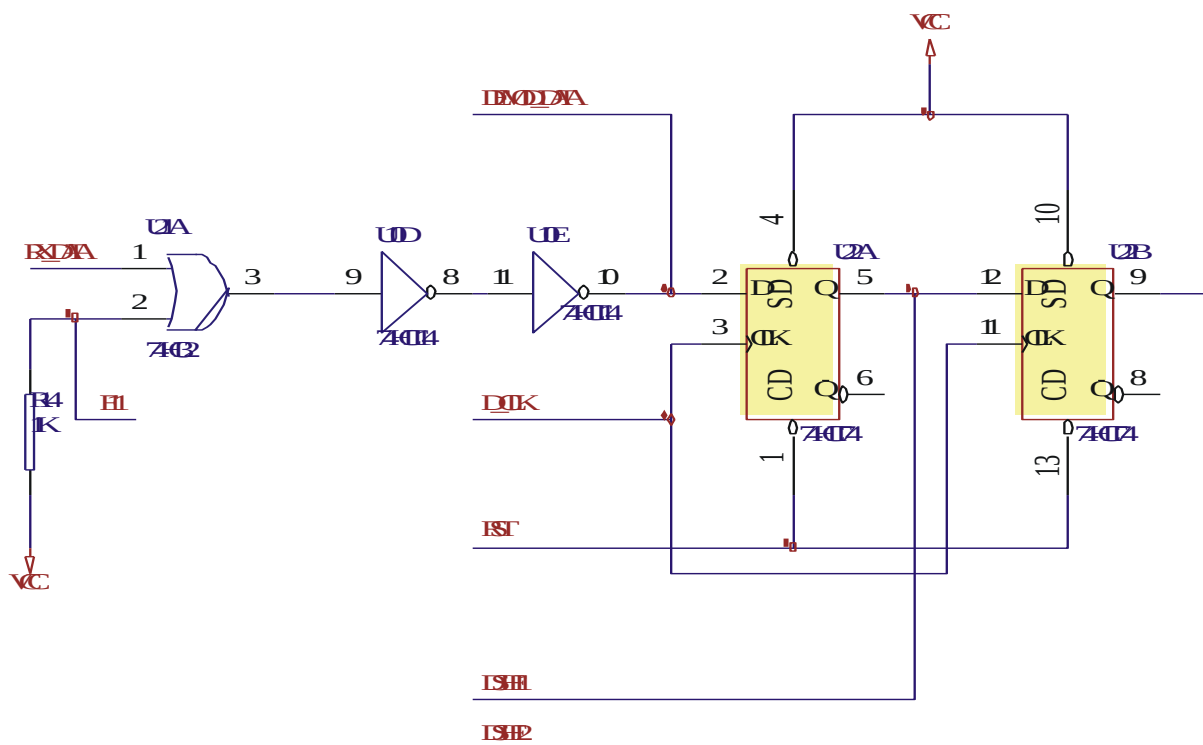




Action and Effect

The predicted data is integrated through the integrator circuit. When we put the 5 of **SF2** (ADCL-07) in Switch Fault section to ON position, the feedback resistor P2 is opened, due to which the integration will not take place properly and effect in quantization error block can be observed as in Experiment No. 3.

6) Effect of Switch Fault 6 (DCL-DPCM) In DPCM Demodulation Section



DESCRIPTION

This circuit gives us the DPCM demodulated data.

PROCEDURE

First keep switch **6 (SF2)** to OFF position. Observe the waveform at the given output test point of the adder in the DPCM DEMODULATOR section.

1. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
2. Keep the clock frequency at 512 KHz, by changing the jumper position in the clock generator section.
3. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.

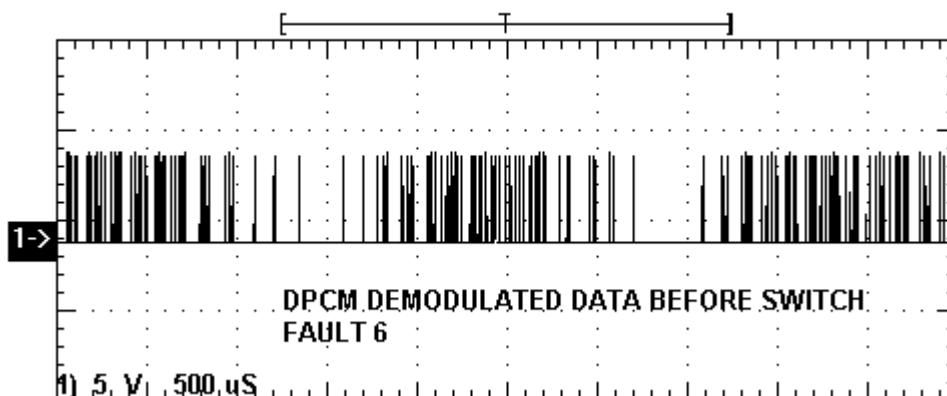
DPCM modulation

4. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect the **OUT** post of buffer to the **IN** DPCM modulator section.
6. Observe the sample output at the given test point. The input signal is sampled at the clock frequency of 16 KHz.
7. Observe the linear predictor output at the **PREDICTED OUT** post of the linear predictor in the DPCM modulator section.
8. Observe the differential pulse code modulated data (DPCM) at the **OUT** post of the DPCM modulator section.

DPCM demodulation

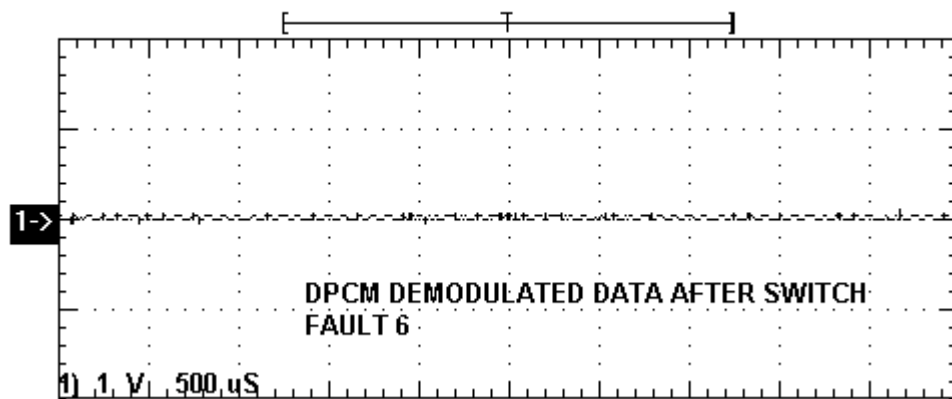
9. Connect the DPCM modulated data from the **OUT** post of the DPCM modulator to the **IN** post of the DPCM demodulator.
10. Observe the demodulated data at the summation out in DPCM demodulation section.

Waveform before Switch Fault



Now put the switch **6 (SF2)** in the Switch Fault to ON position and observe its effect.

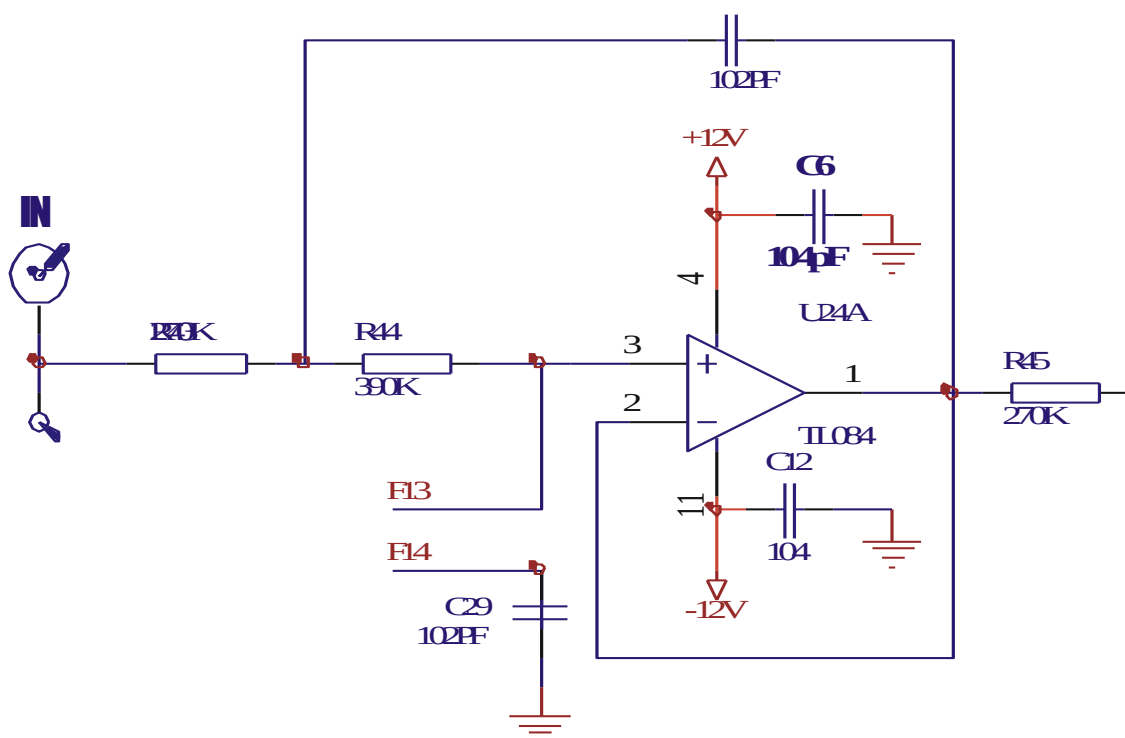
Waveform after Switch Fault



Action and Effect

This circuit gives us the DPCM demodulated data. When we put the **6** of **SF2** (DCL-DPCM) in Switch Fault section to ON position, the predictor section is isolated from the adder. Therefore the demodulated data at summation out test point in DPCM demodulated section is not derived.

7) Effect of Switch Fault 7 (DCL-DPCM) In Filter



DESCRIPTION

This section filters the integrated output of the integrator.

PROCEDURE

First keep switch **7 (SF2)** to OFF position. Observe at filter out.

1. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
2. Keep the clock frequency at 512 KHz, by changing the jumper position in the clock generator section.
3. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.

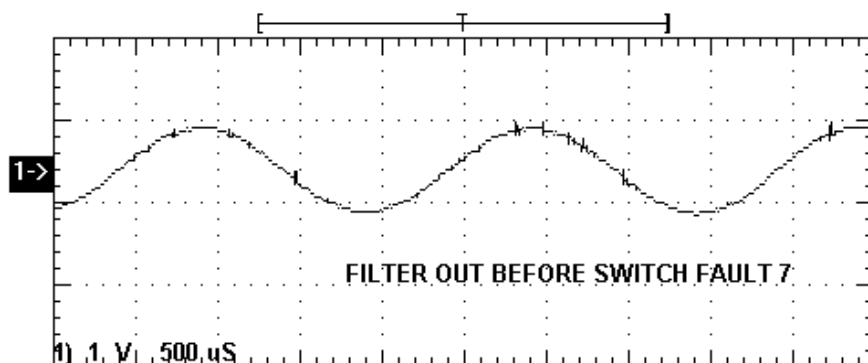
DPCM modulation

4. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect the **OUT** post of buffer to the **IN** DPCM modulator section.
6. Observe the sample output at the given test point. The input signal is sampled at the clock frequency of 16 KHz.
7. Observe the linear predictor output at the **PREDICTED OUT** post of the linear predictor in the DPCM modulator section.
8. Observe the differential pulse code modulated data (DPCM) at the **OUT** post of the DPCM modulator section.

DPCM demodulation

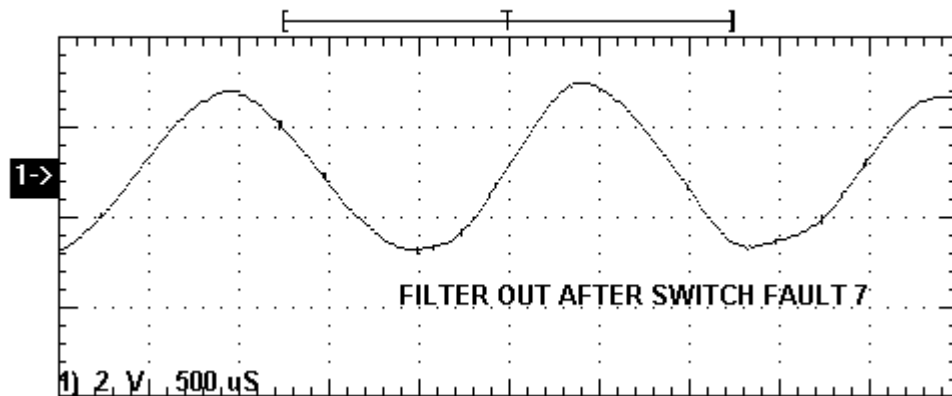
9. Connect the DPCM modulated data from the **OUT** post of the DPCM modulator to the **IN** post of the DPCM demodulator.
10. Observe the demodulated data at the given test point.
11. Observe the integrated demodulated data at the **OUT** post of the DPCM demodulator.
12. Connect the demodulated data from the **OUT** post of the DPCM demodulator to the **IN** post of the low-pass filter.
13. Observe the reconstructed signal at the **OUT** post of the filter.

Waveform before Switch Fault



Now put the switch **7 (SF2)** in the Switch Fault to ON position and observe its effect.

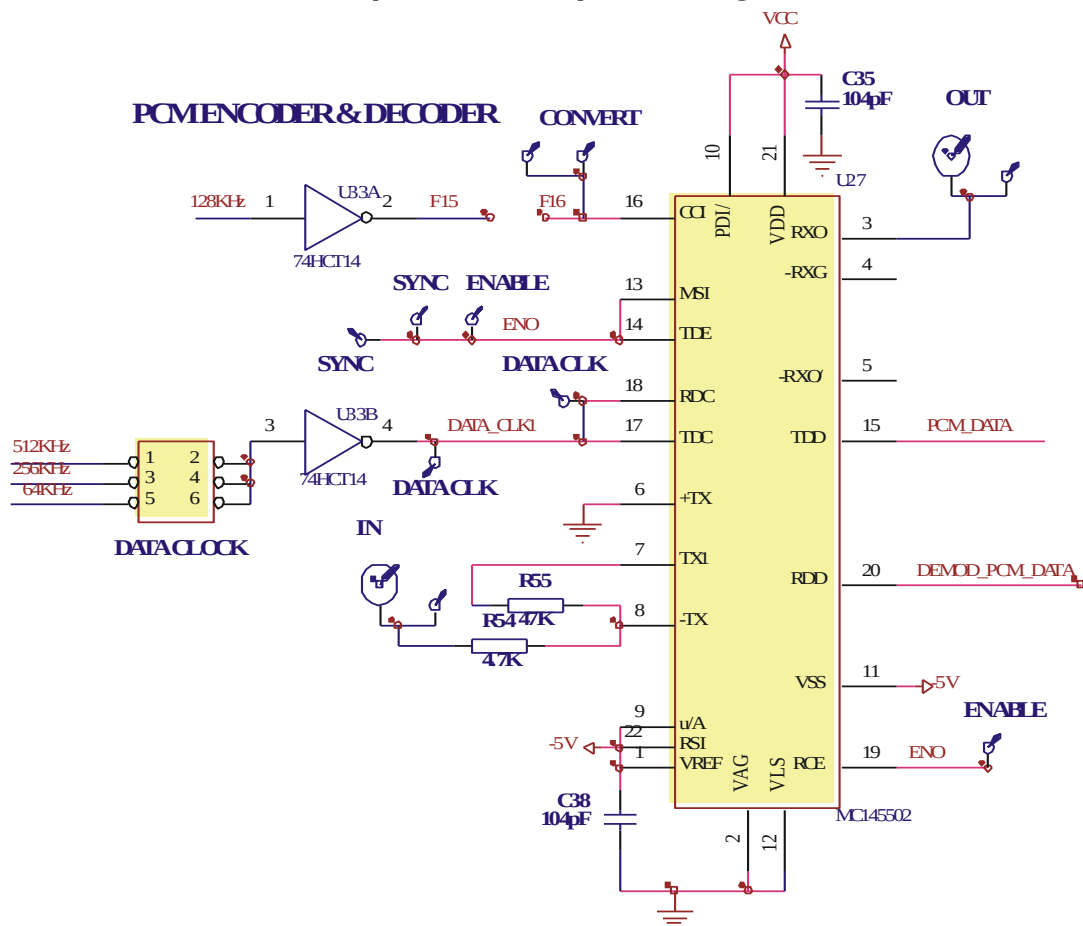
Waveform after Switch Fault



Action And Effect

This section filters the integrated output of the integrator. When we put the **7** of **SF2** (ADCL-07) in Switch Fault section to ON position, the cap C29 is in open state due to which the filtration is not proper.

8) Effect of Switch Fault 8 (DCL-DPCM) in data generation section:



DESCRIPTION

This circuit converts the analog signal into PCM data and vice versa. The convert clock input frequency is compared with the master synchronization input if frequency 8KHz and prescale divided to produce the internal sequencing clock.

PROCEDURE

First keep switch **8 (SF2)** to OFF position. Observe the waveform at SDATA port.

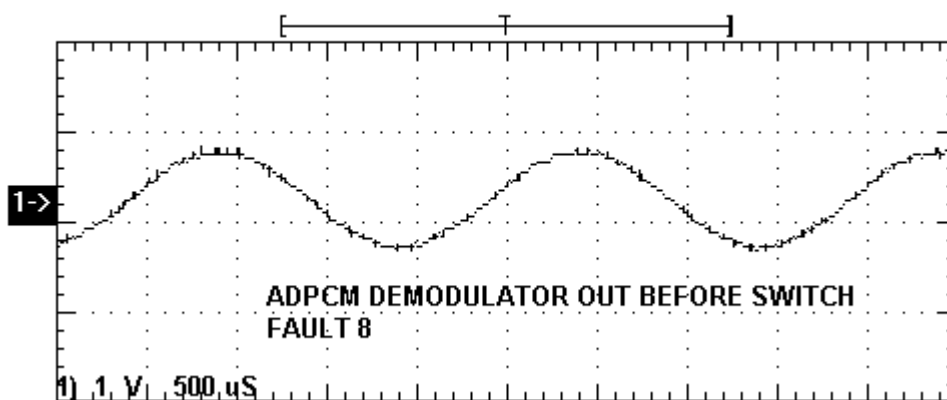
ADPCM modulation

1. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
2. Keep the clock frequency at 512 KHz, by changing the jumper position in the clock generator section.
3. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.
4. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect the **OUT** post of buffer to the **IN** DPCM modulator section.
6. Observe the PCM data at the given test point in the ADPCM modulator section.
7. Observe the ADPCM data at the **OUT** post of ADPCM modulator section.

ADPCM demodulation:

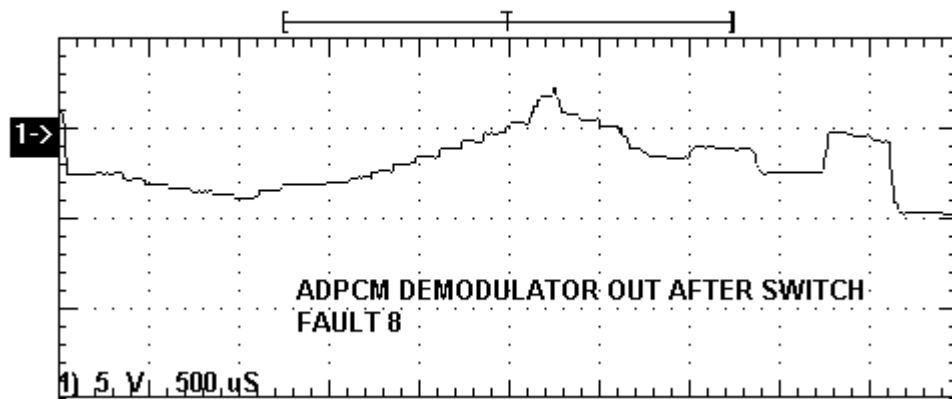
8. Connect the ADPCM demodulated data from the **OUT** post of the ADPCM modulator to the **IN** post of the ADPCM demodulator.
9. Observe the demodulated PCM data at the provided test point. Observe the reconstructed sine wave at the **OUT** post of the ADPCM demodulator.

Waveform before Switch Fault



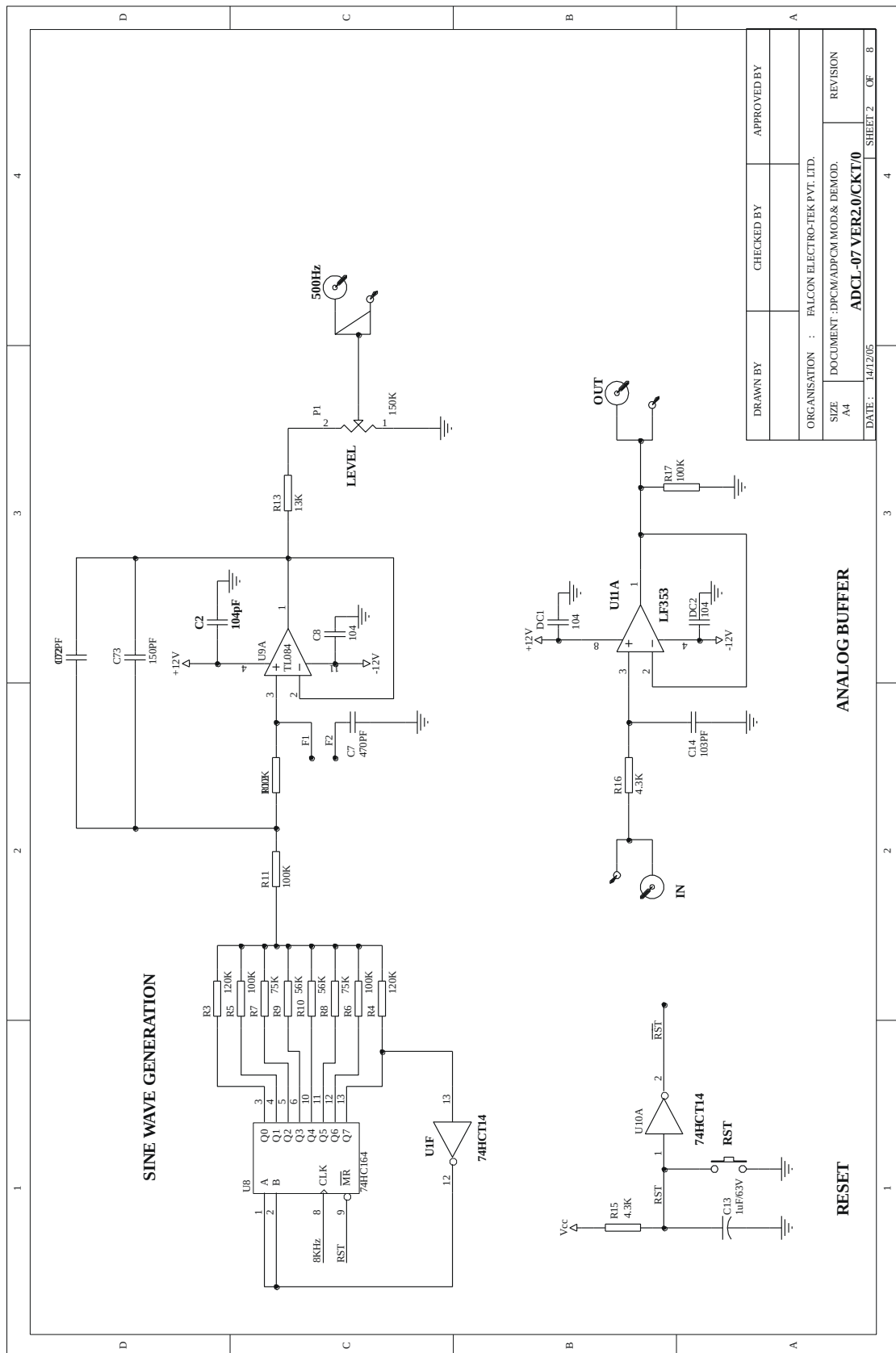
Now put the switch **8 (SF2)** in the Switch Fault to ON position and observe its effect.

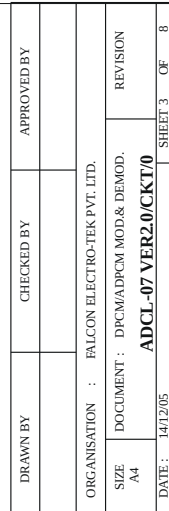
Waveform after Switch Fault

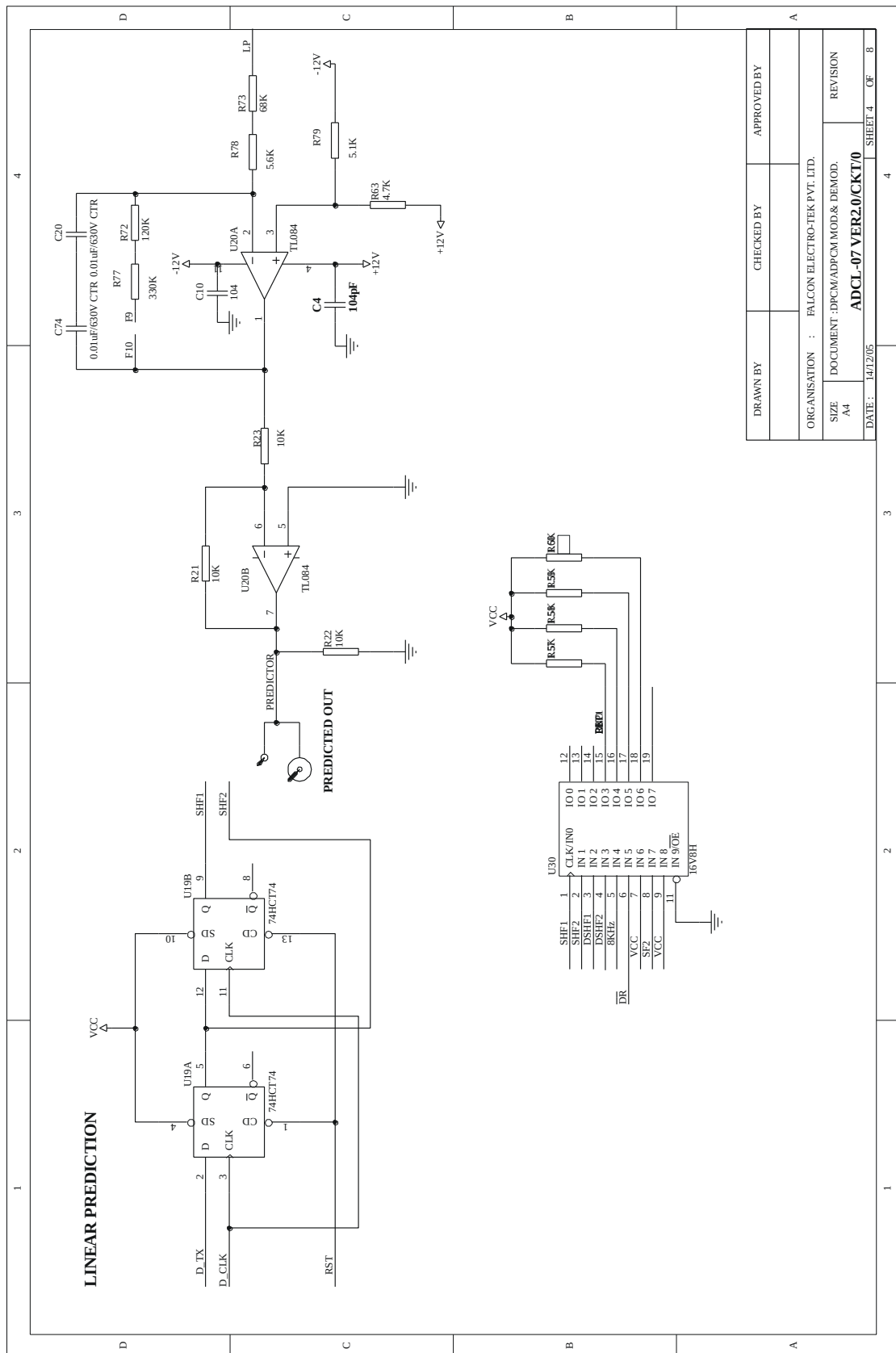


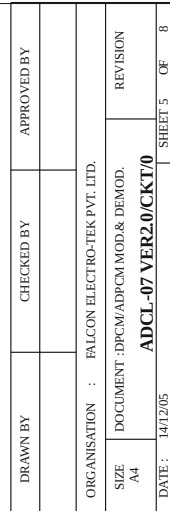
Action and Effect

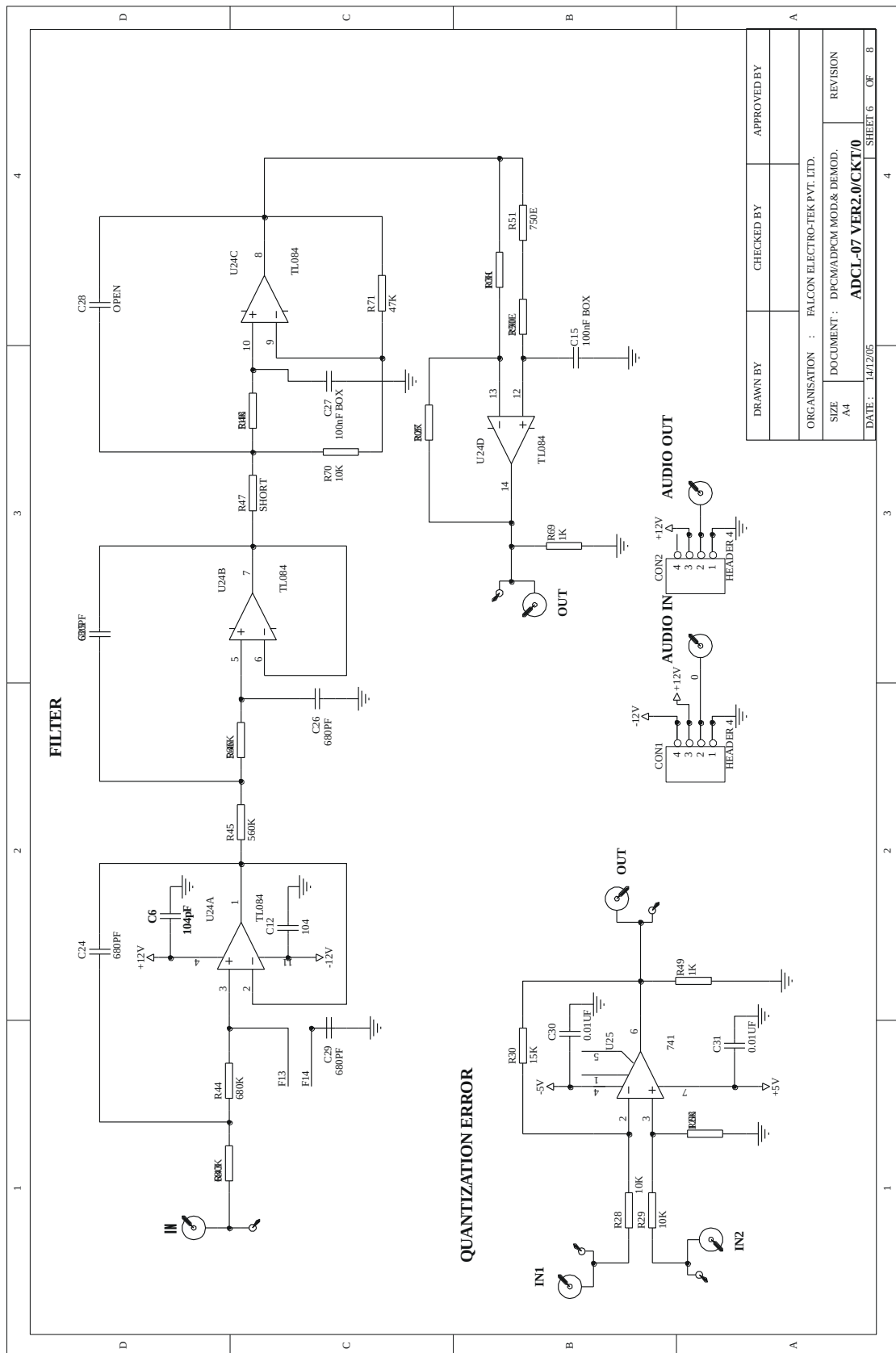
When we put the **8** of **SF2** (DCL-DPCM) in Switch Fault section to ON position, the convert clock of 128KHz going to the chip is removed, due to which unwanted signal will appear at the output or signal will completely disappear.

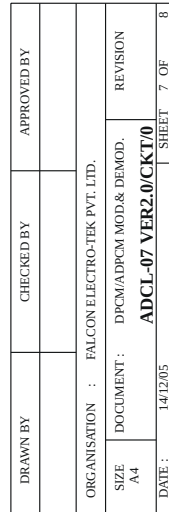


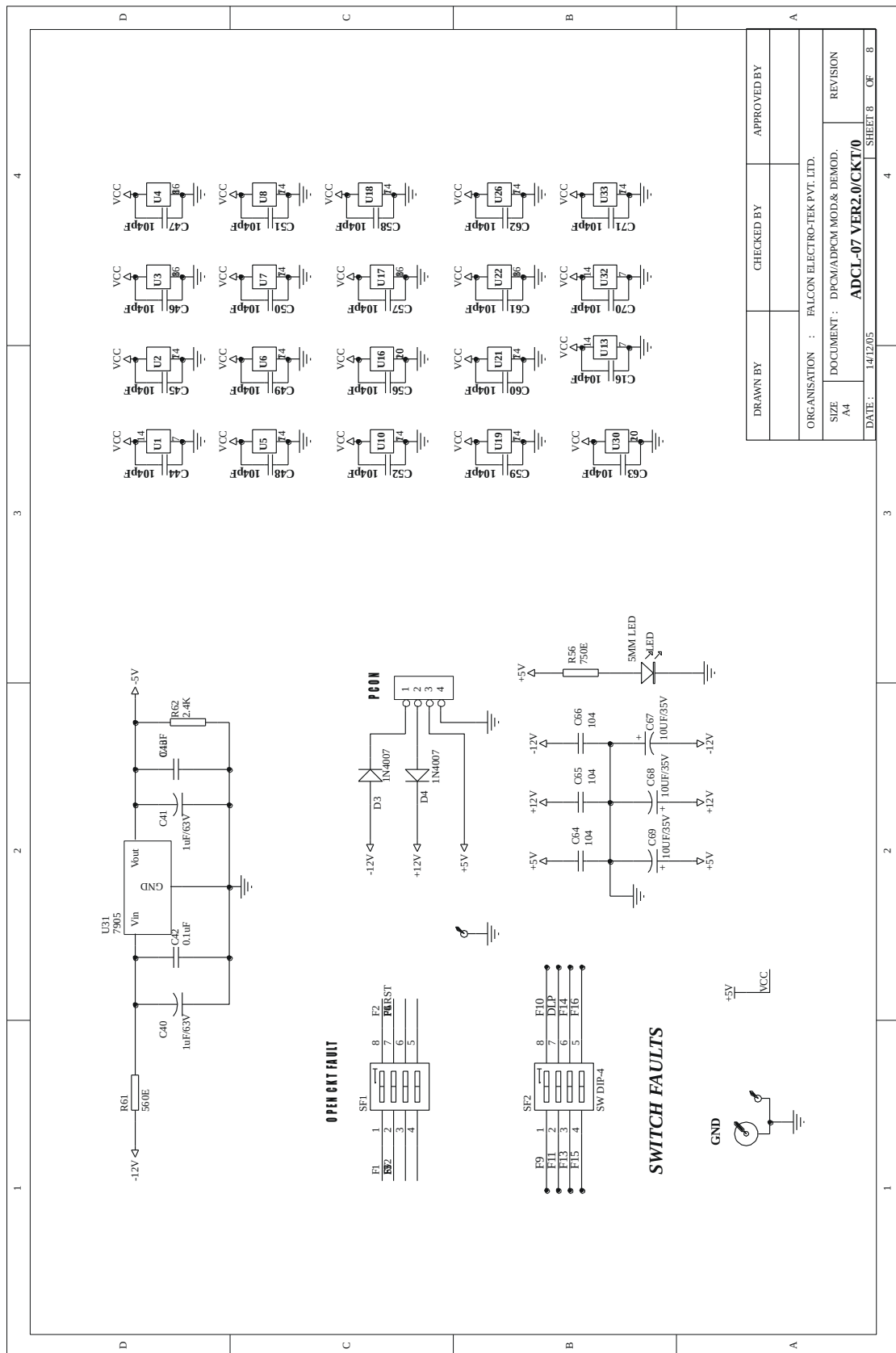












EXPERIMENT NO. 1

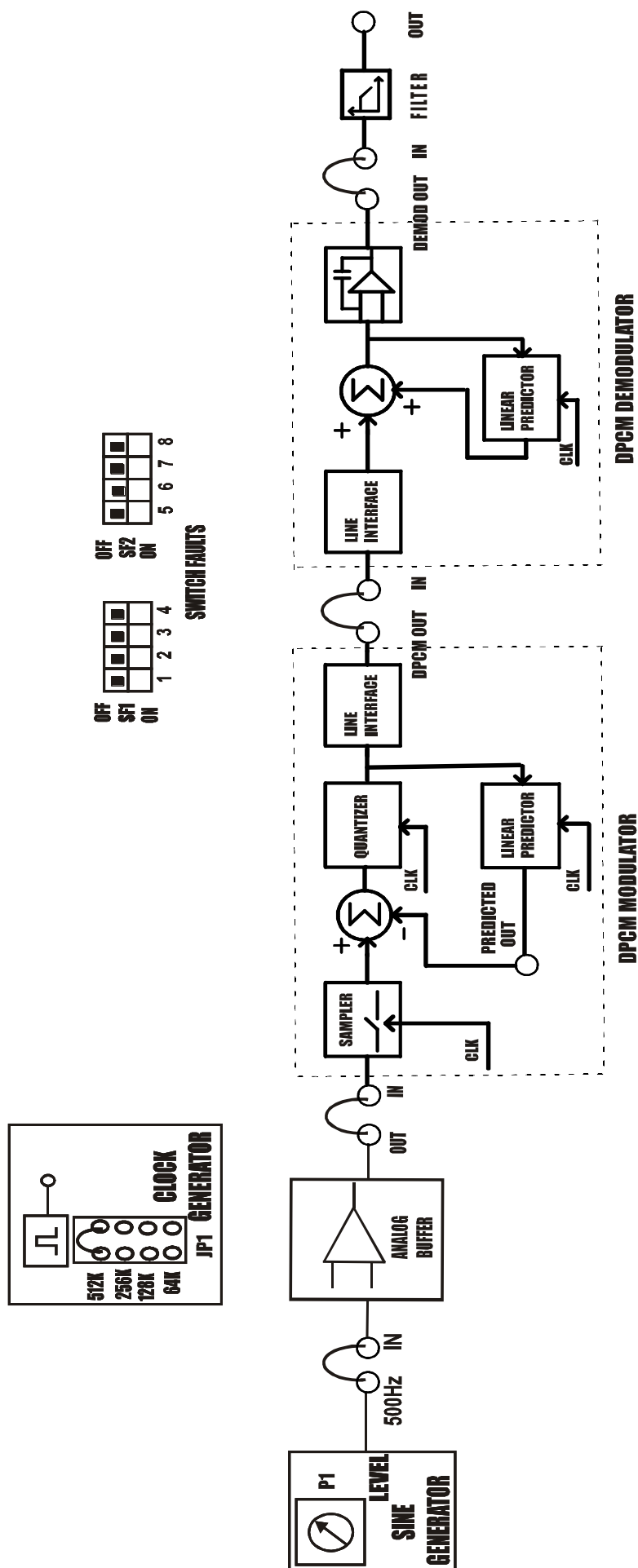


FIG. 1 BLOCK DIAGRAM FOR STUDY OF DIFFERENTIAL PULSE CODE MODULATION AND DEMODULATION

EXPERIMENT NO: 1

NAME:

Differential Pulse Code Modulation and Demodulation

OBJECTIVE:

Study of differential pulse code modulation technique

THEORY:

DPCM is a good way to reduce the bit rate for voice transmission. However it causes some other problems that deals with voice quality. DPCM quantizes and encodes the difference between a previous sample input signal and a current sample input signal. DPCM quantizes the difference signal using uniform quantization. Uniform quantization generates an SNR that is small for small input sample signals and large for large input sample signals. Therefore, the voice quality is better at higher signals.

The first part of DPCM works exactly like PCM (that is why it is called differential PCM). The input signal is sampled at a constant sampling frequency (more than the input frequency). Then these samples are modulated. At this point, the DPCM process takes over. The sampled input signals are stored in what is called a predictor. The predictor takes the stored sample signal and sends it through a differentiator. The differentiator compares the previous sample signal and sends its difference to the quantizing and coding phase of PCM (this phase can be a uniform quantizing or companding with A-law or μ -law). After quantizing and coding, the difference signal is transmitted to its final destination. At the receiving end of the network, everything is reversed. First the difference signal is decoded and dequantized. This difference is added to the sample signal stored in the predictor and send through a low-pass filter that reconstructs the original input signal.

EQUIPMENTS:

- DCL-DPCM Kit
- Connecting Chords
- Power Supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

1. Refer to the block diagram (Fig.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
3. Keep the clock frequency at 512 KHz, by changing the jumper position of JP1 in the clock generator section.

4. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.

DPCM modulation:

- Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
5. Connect **OUT** post of Analog Buffer to **IN** post of DPCM modulator section.
6. Observe the sample output at the given test point. The input signal is sampled at the clock frequency of 16 KHz.
7. Observe the linear predictor output at the **PREDICTED OUT** post of the linear predictor in the DPCM modulator section.
8. Observe the differential pulse code modulated data (DPCM) at the **DPCM OUT** post of the DPCM modulator section.
9. Observe the DPCM data at **DPCM OUT** post by varying input signal from 0 to 2V.

DPCM demodulation:

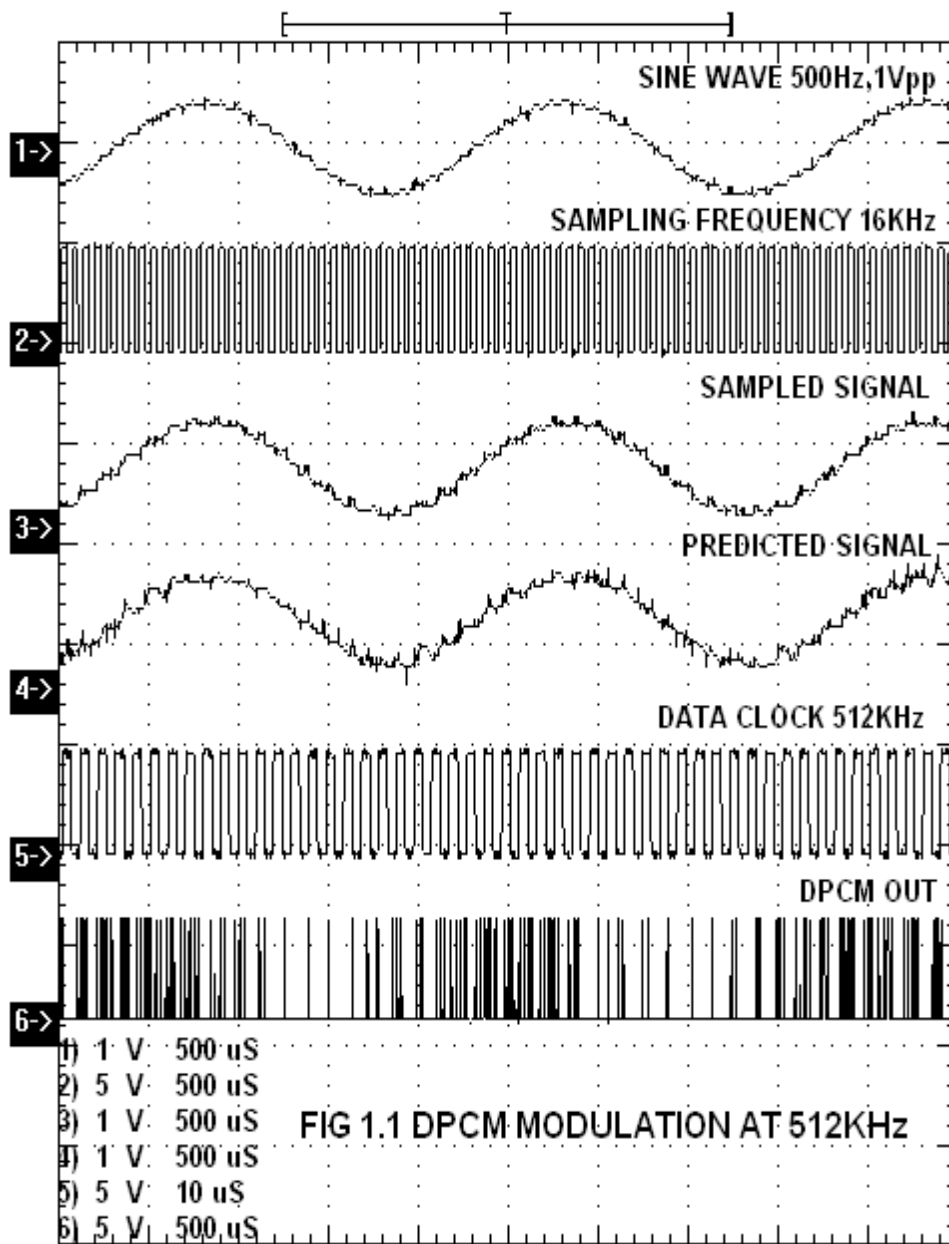
10. Connect the DPCM modulated data from the **DPCM OUT** post of the DPCM modulator to the **IN** post of the DPCM demodulator.
11. Observe the demodulated data at the output of summation block.
12. Observe the integrated demodulated data at the **DEMOD OUT** post of the DPCM demodulator.
13. Connect the demodulated data from the **DEMOD OUT** post of the DPCM demodulator to the **IN** post of the low-pass filter.
14. Observe the reconstructed signal at the **OUT** post of the filter. Use **RST** switch for clear observation of output.
15. Now, simultaneously reduce the clock frequencies from **512 KHz** to **256 KHz**, **128 KHz** and **64 KHz** by changing the jumper position of JP1 and observe the difference in the DPCM modulated and demodulated data. As the frequency of clock decreases, DPCM demodulated data at **DEMOD OUT** becomes distorted.
16. Observe various waveforms as mentioned below (Fig. 1.2).

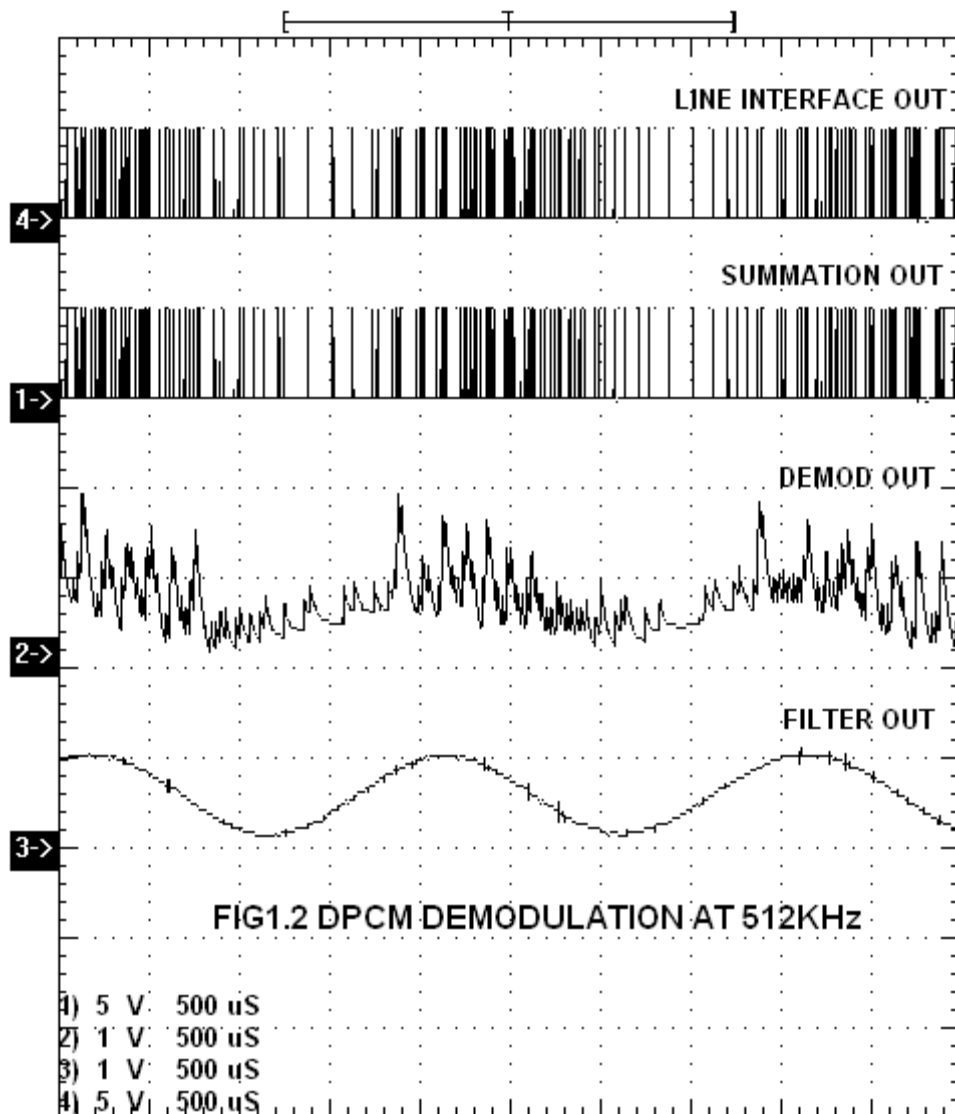
OBSERVATION:

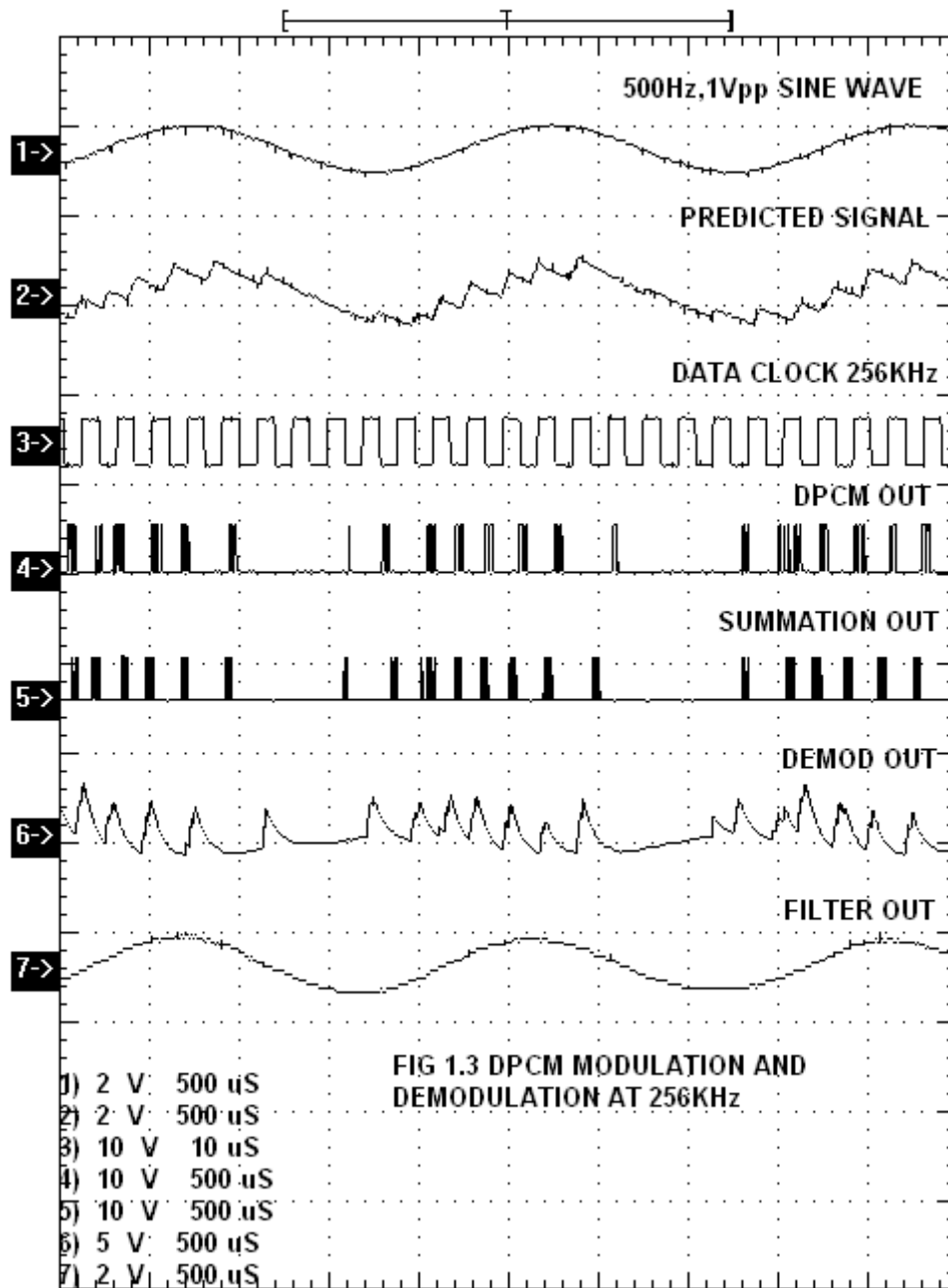
ON KIT DCL-DPCM

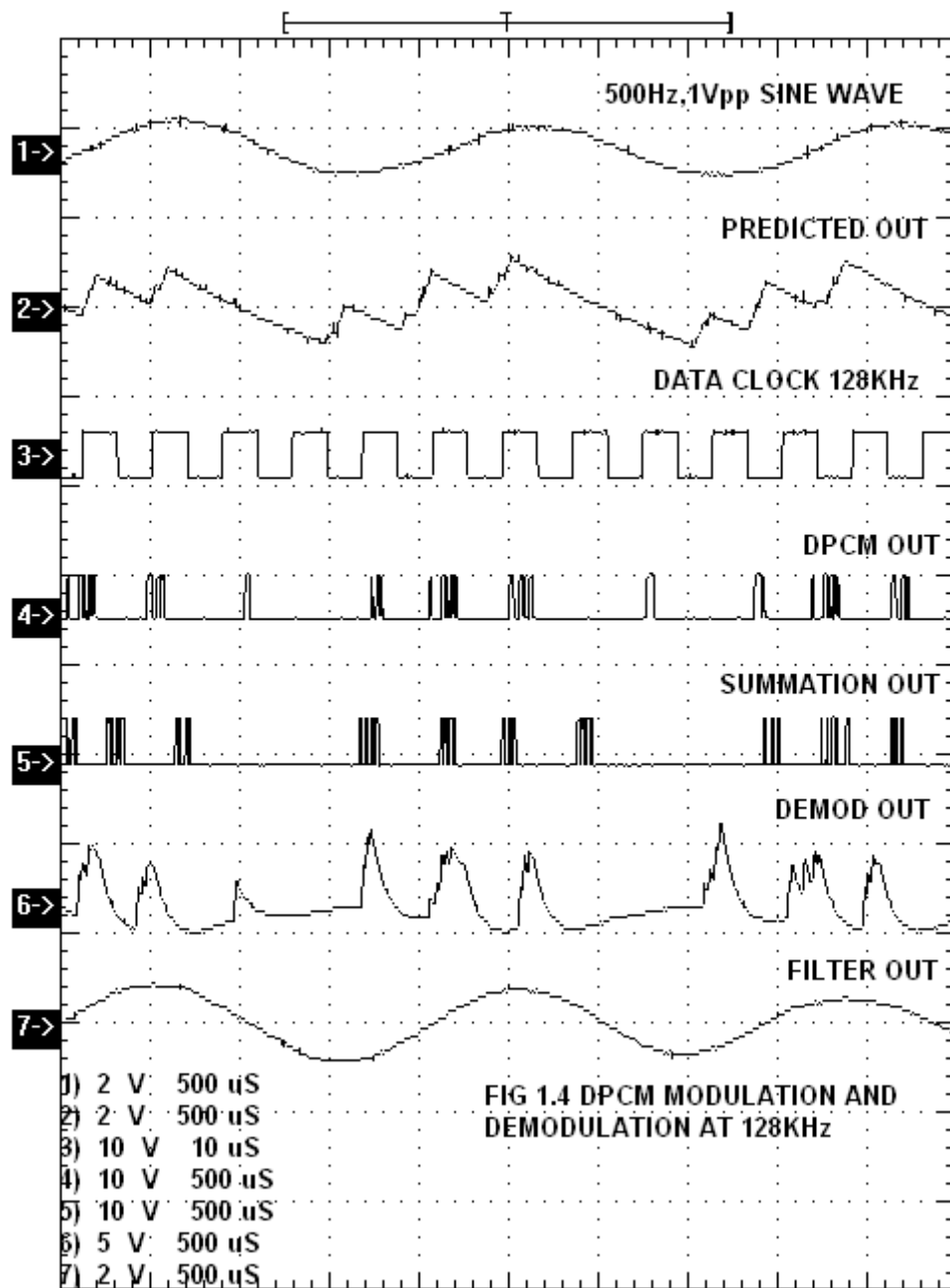
Observe the following waveforms on the oscilloscope and plot on the paper.

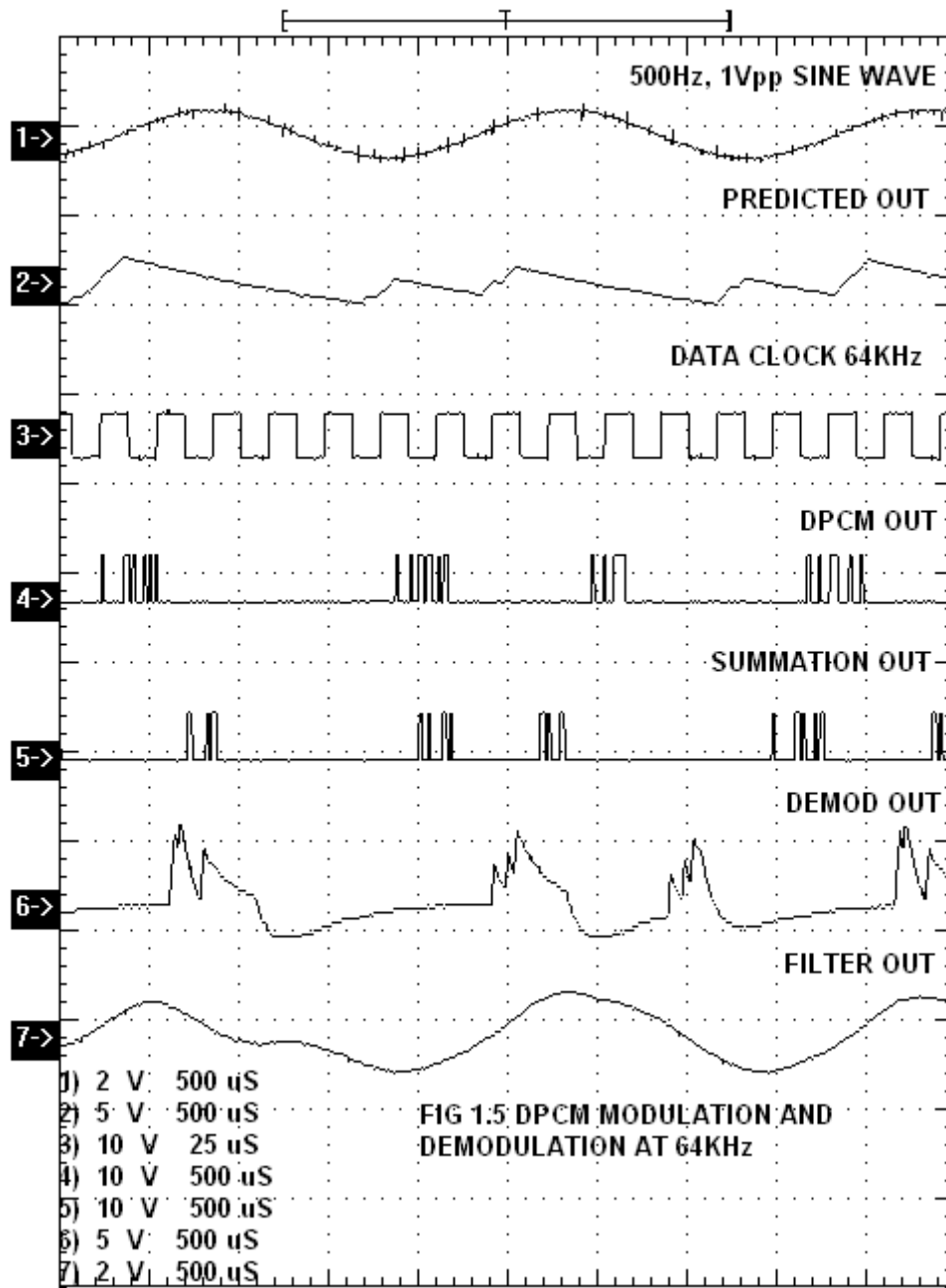
- 500Hz, 1Vpp input sine wave.
- Sampled out at the provided test point **SAMPLER OUT**.
- Linear predictor out at **PREDICTED OUT** post.
- DPCM data at **DPCM OUT** post.
- Line interface out at the given output test point of line interface block in DPCM demodulator.
- Demodulated DPCM data at the output test point of summation block in DPCM demodulator.
- Integrated demodulated data at the **DEMOD OUT** post of the DPCM demodulator.
- Reconstructed sine wave at the **OUT** post of the filter.
- Observe the data at different clock rates.



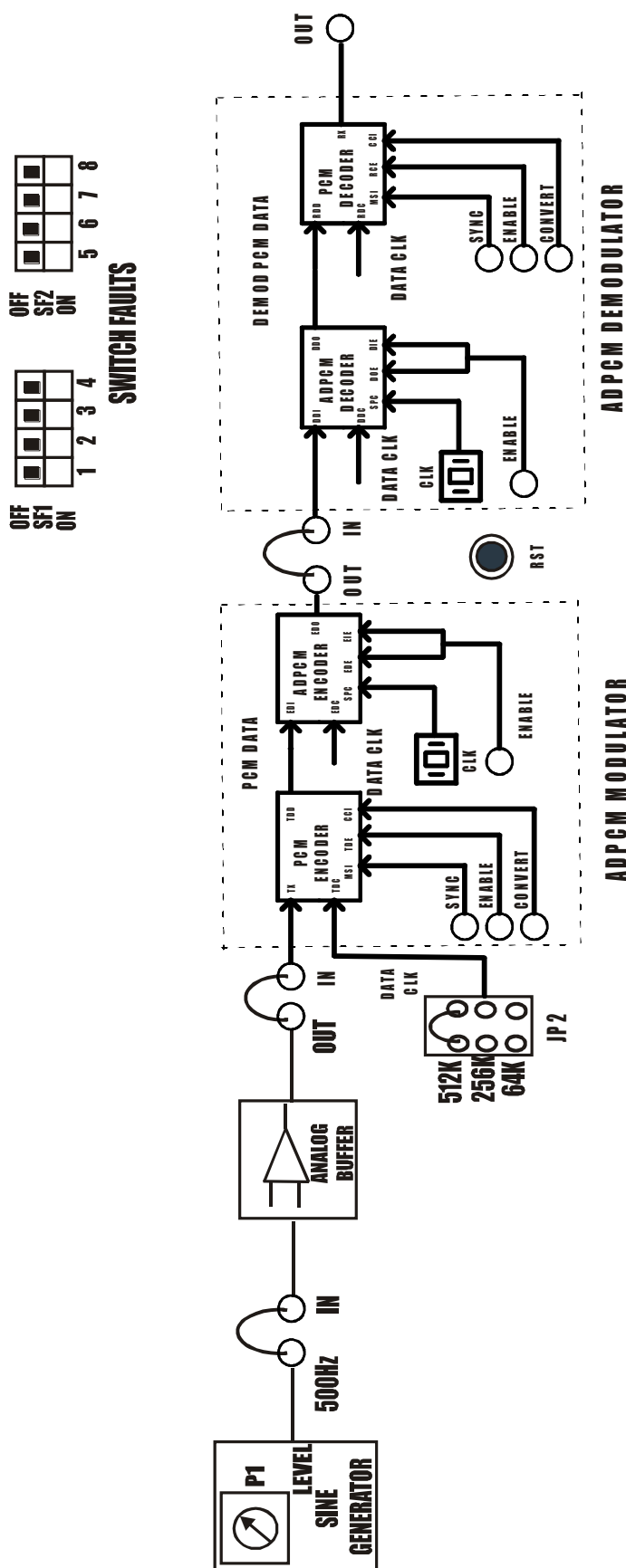








EXPERIMENT NO. 2



EXPERIMENT NO: 2

NAME:

Adaptive Differential Pulse Code Modulation And Demodulation

OBJECTIVE:

To study the adaptive differential pulse code modulation technique

THEORY:

Since most of the signals generated by the human voice are small. Voice quality needs to focus on small signals. To solve this problem, adaptive DPCM is developed. Adaptive DPCM adapts the quantization levels of the difference signal that generated at the time of DPCM process. When the difference signal is low, ADPCM increases the size of the quantization levels. If the difference signal is high, ADPCM decreases the size of the quantization levels. So, ADPCM adapts the quantization level to the size of the input difference signal. This generates an SNR that is uniform through out the dynamic range of the difference signal. Using ADPCM reduces bit rate of voice transmission.

CODEC IC is used for analog to digital conversion. The whole process takes place inside the CODEC. Present techniques of voice communication standards such as A-law / μ -law companded PCM voice coding at 64kbps. If the amplitude of the signal is small; quantization levels have to be closely spaced. This gives proper resolution. But if the signal amplitude is large than this fine resolution will result in increasing the number of code bits. The analog input is fed into the CODEC; the output is in digital form. The digital data output is in Pulse Code Modulated form. The CODEC chips used exhibit both A-law and μ -law companding techniques. Here, we select this A-law and μ -law companding with the help of a switch.

The PCM data is fed to the ADPCM CODEC IC. This CODEC is used to reduce the data rate required to transmit a PCM encoded analog / voice signal while maintaining the voice fidelity and intelligibility of the PCM signal. It also uses a filter to attempt to predict the next PCM input value based on previous PCM input values. The error between the predicted and the true PCM input value is the information that is sent to the other end of the line. The adaptive filter adapts to the statistics of the signals presented to it. Here also the whole process takes place inside the CODEC.

CODEC IC is used for ADPCM demodulation also. The ADPCM data is fed into this CODEC along with clocks for synchronization. Demodulated PCM data is derived. This PCM data is fed into the PCM CODEC filter IC. The received digital data is again converted into analog form by the chip.

EQUIPMENTS:

- DCL-DPCM Kit
- Connecting Chords
- Power Supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

ADPCM modulation:

1. Refer to the block diagram (Fig.2) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
3. Keep the clock frequency at 512 KHz, by changing the jumper position of JP2 in the clock generator section.
4. Keep the amplitude of the onboard sine wave, of frequency 500Hz to 1Vpp.
5. Connect the 500Hz sine wave to the **IN** post of Analog Buffer.
6. Connect the **OUT** post of buffer to the **IN** post of ADPCM modulator section.
7. Observe the PCM data at the given test point in the ADPCM modulator section.
8. Observe the ADPCM data in DUAL mode at the **OUT** post of ADPCM modulator section with sync pulse and with input sine wave.

ADPCM demodulation:

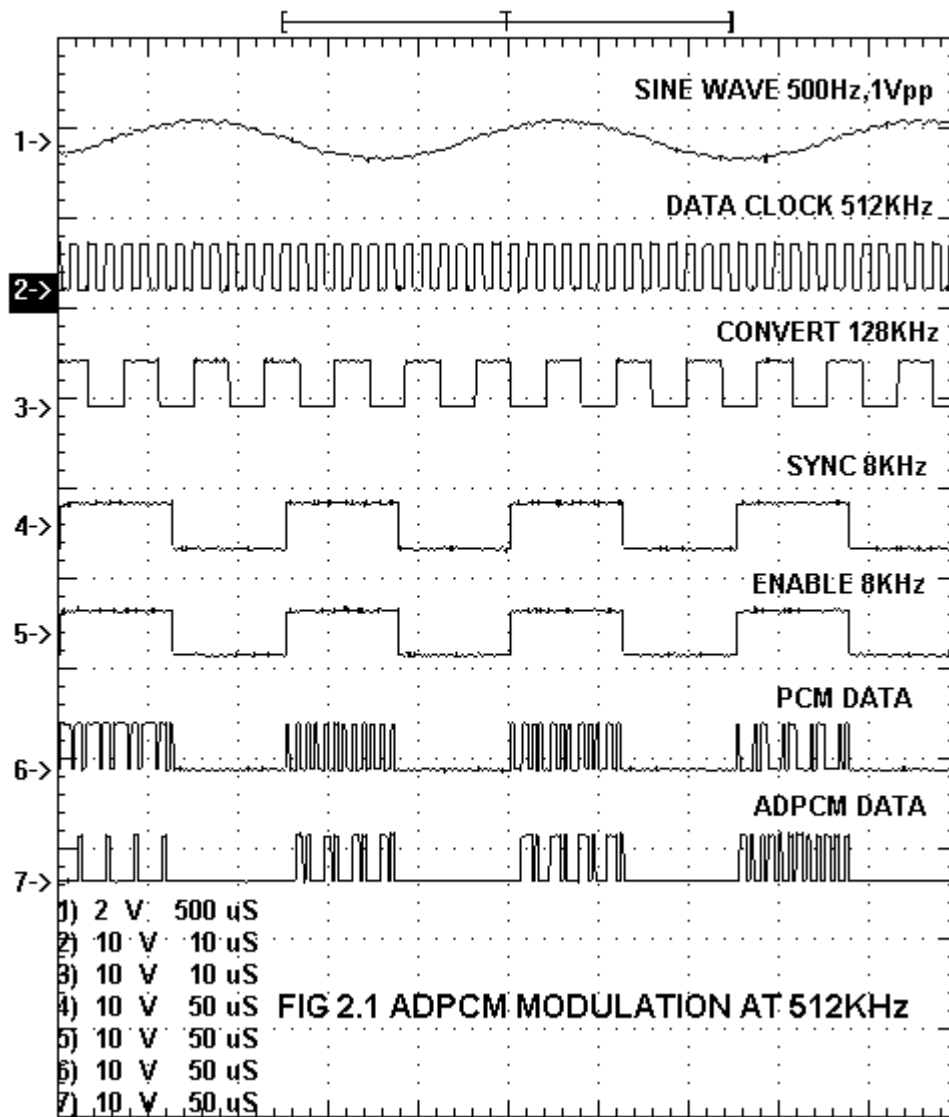
9. Connect the ADPCM demodulated data from the **OUT** post of the ADPCM modulator to the **IN** post of the ADPCM demodulator.
10. Observe the demodulated PCM data at the provided test point. Observe the reconstructed sine wave at the **OUT** post of the ADPCM demodulator. Use **RST** switch for clear observation of output.
11. Now, simultaneously reduce the clock frequencies from 512 KHz to 256 KHz and 64 KHz by changing the jumper position of JP2 and observe the difference in the data and the reconstructed sine wave. As the frequency of clock decreases, DPCM modulated data at filter **OUT** becomes distorted.
12. Observe various waveforms as mentioned below.

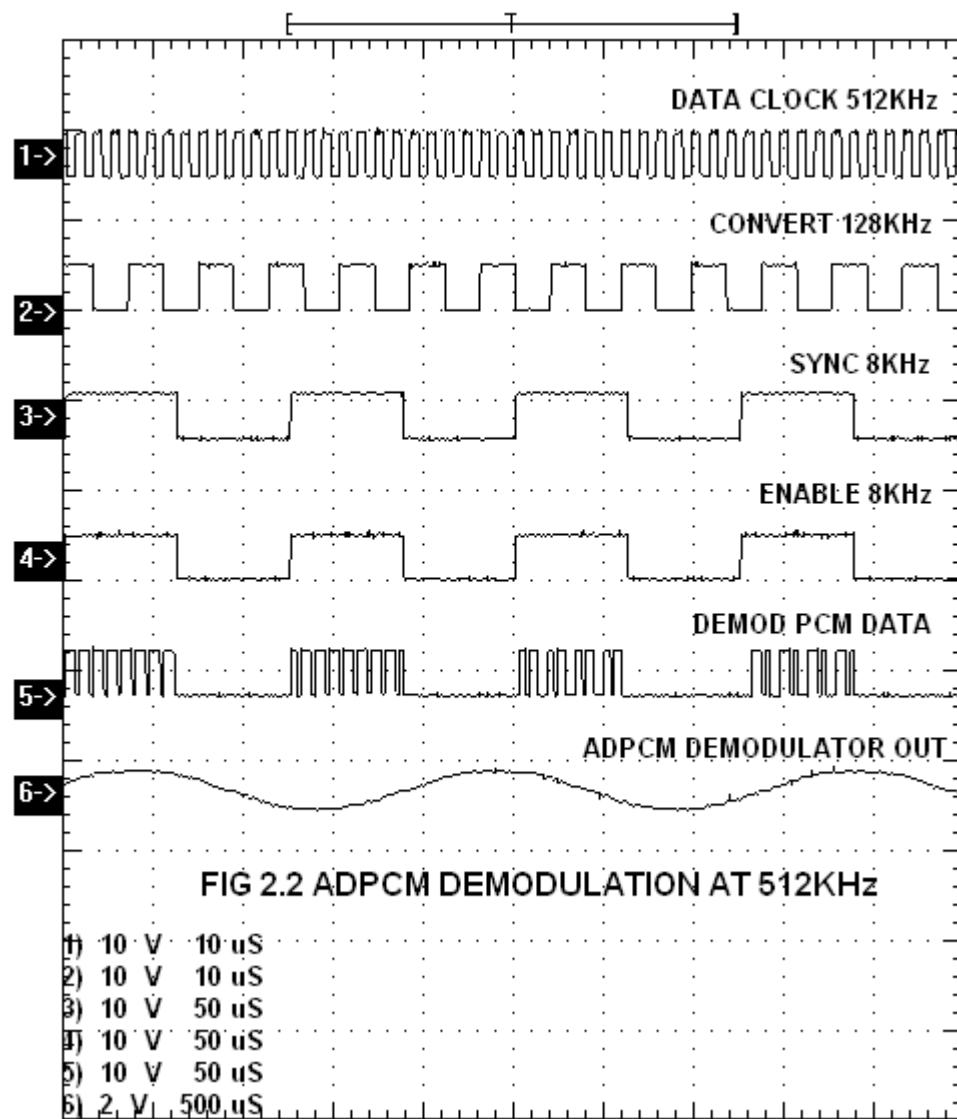
OBSERVATION:

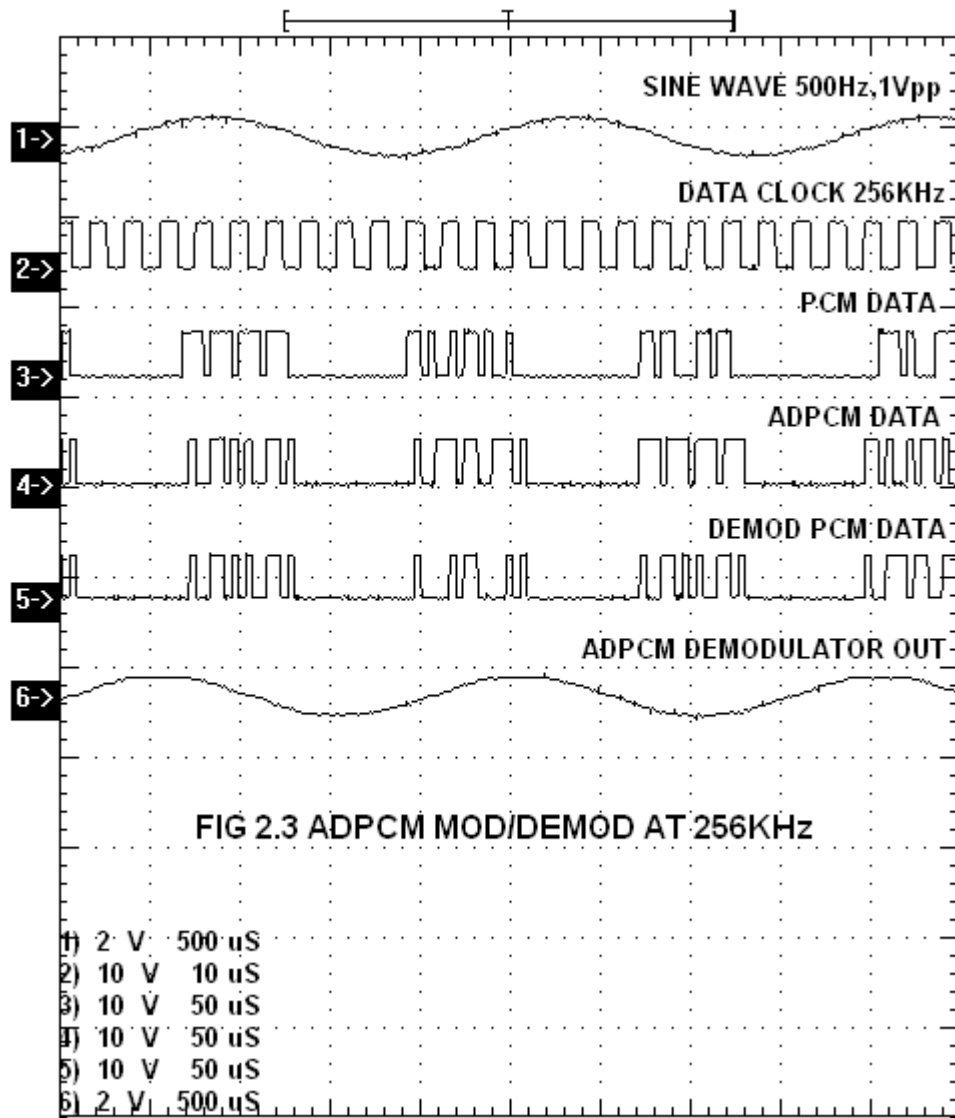
ON KIT **DCL-DPCM**

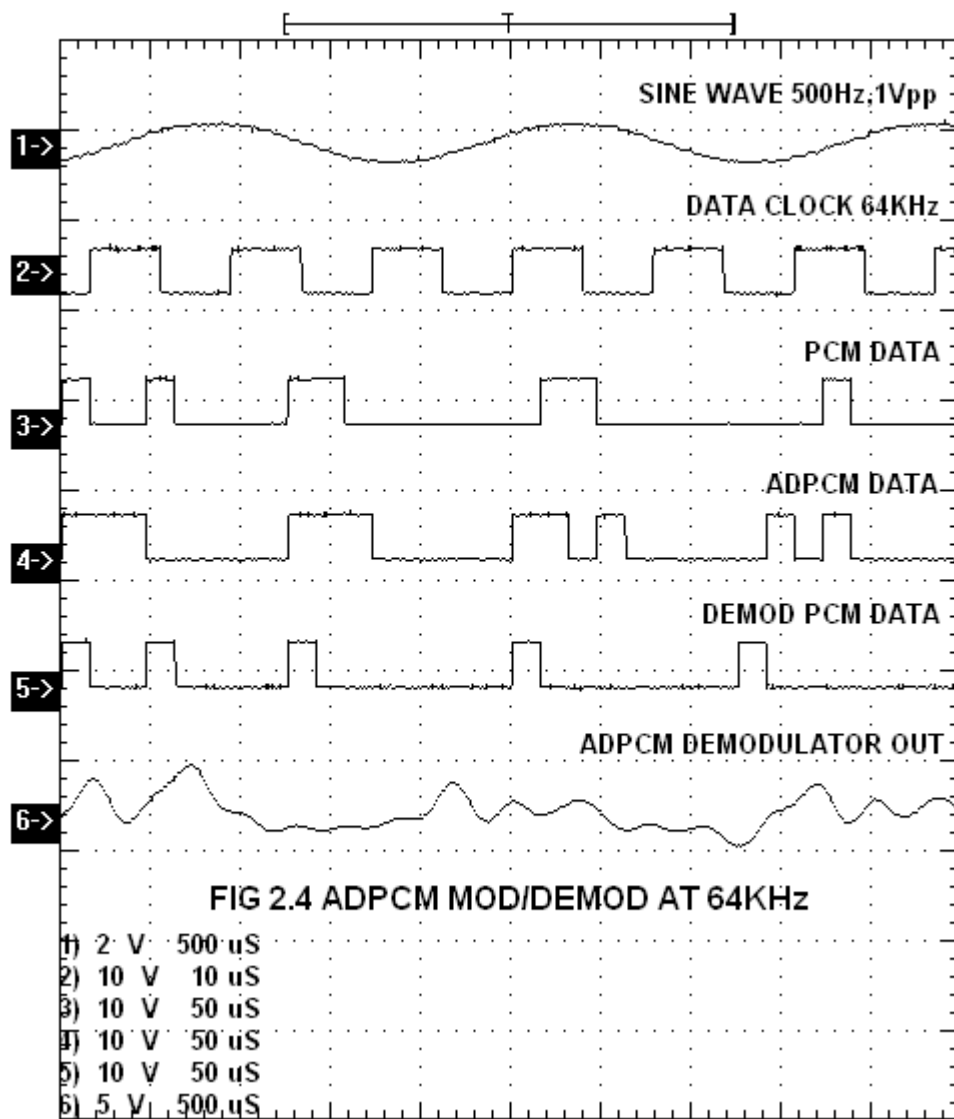
Observe the following waveforms on oscilloscope and plot on the paper.

- Input sine wave 500Hz, 1Vpp.
- PCM data at different frequencies at the given test point on the ADPCM modulator section.
- ADPCM data at different frequencies the **OUT** post of the ADPCM modulator section.
- Demodulated PCM data at different frequencies at the given test point of ADPCM demodulator section.
- Reconstructed sine wave at different frequencies at the **OUT** post of the ADPCM demodulator section.









EXPERIMENT NO. 3

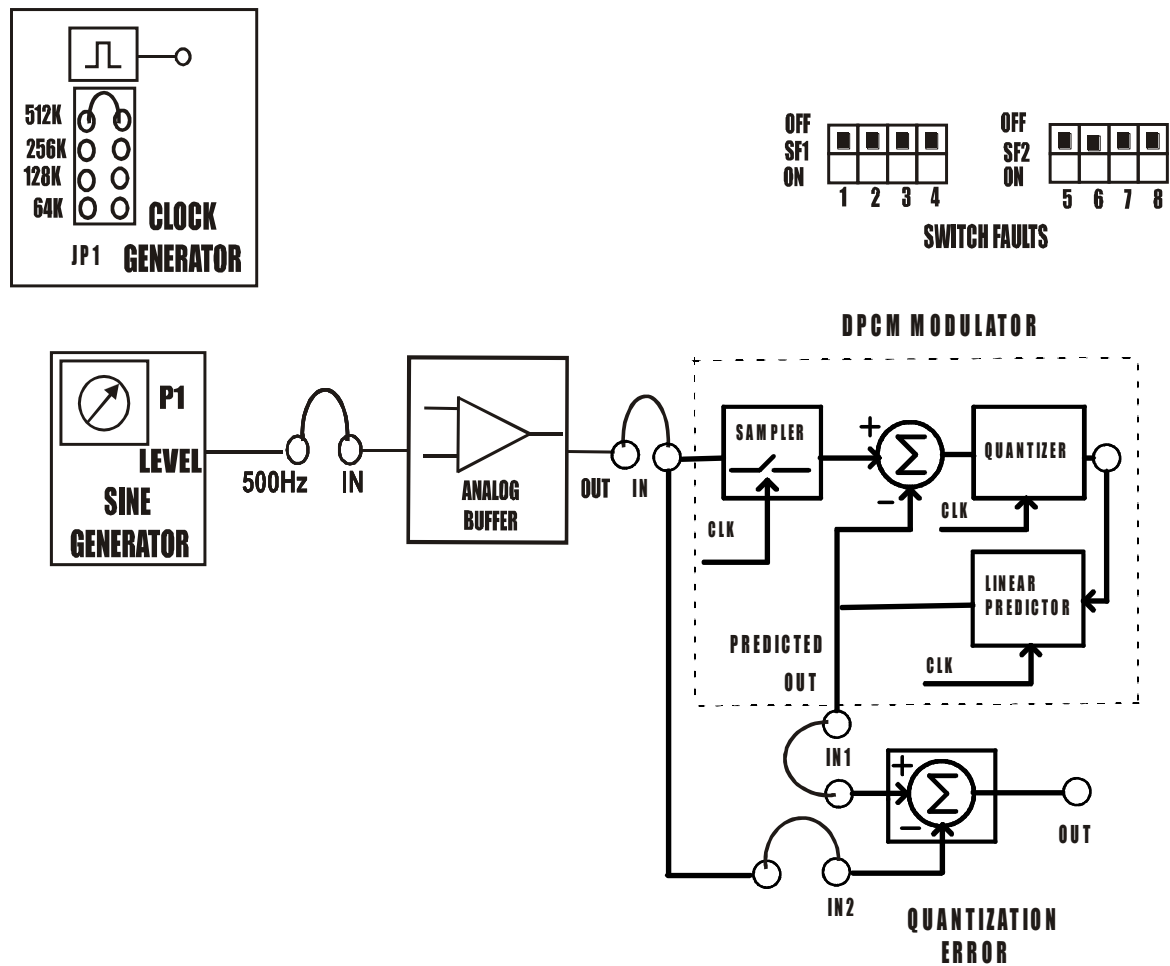


FIG. 3 BLOCK DIAGRAM FOR THE STUDY OF QUANTIZATION ERROR

EXPERIMENT NO: 3

NAME:

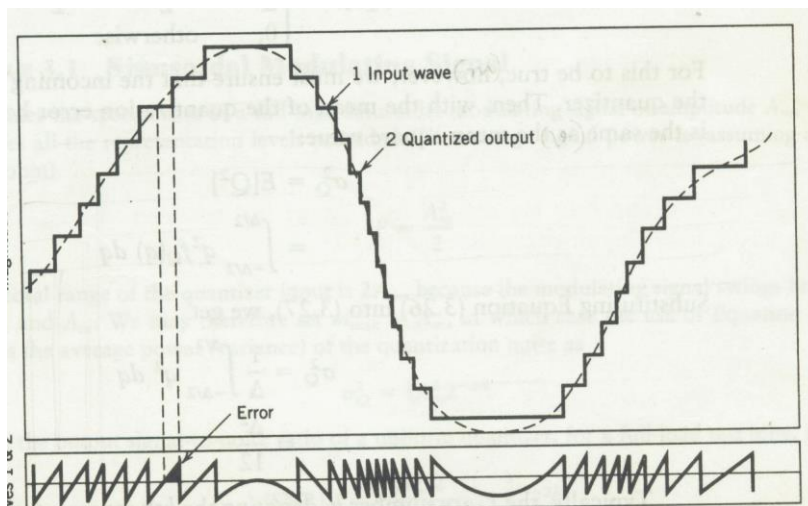
Quantization Error

OBJECTIVE:

Study of quantization error in DPCM modulation

THEORY:

The use of quantization introduces an error defined as the difference between the input signal m and the output signal v . The error is called quantization noise. The figure illustrates a typical variation of the quantization noise as a function of time, assuming the use of a uniform quantizer of the midtread type.



Let the quantizer input m be the sample value of a zero-mean random variable M . (If the input has the nonzero mean, we can always remove it by subtracting the mean from the input and adding it back after quantization.) A quantizer $g(\bullet)$ maps the input random variable M of continuous amplitude into a discrete random variable V ; their respective sample values m and v are related. Let the quantization error be denoted by the random variable Q of sample value q . We may thus write

$$q = m - v \quad (i)$$

Or, correspondingly,

$$Q = M - V \quad (ii)$$

With the input M having zero mean, and the quantizer assumed to be symmetric, it follows that the quantizer output V and therefore the quantization error Q , will also have zero mean. Thus for a partial static characterization of the quantizer in terms of output signal-to- (quantization) noise ratio, we need only find the mean square value of the quantization error Q .

Consider then an input m of continuous amplitude in the range $(-m_{\max}, m_{\max})$. Assuming a uniform quantizer of the midrise type, we find that the step-size of the quantizer is given by

$$\Delta = 2 m_{\max} / L \quad (\text{iii})$$

Where L is the total number of representation levels. For a uniform quantizer, the quantization error Q will have its sample values bounded by $-\Delta/2 \leq q \leq \Delta/2$. If the step size Δ is sufficiently small (i.e., the number of representation levels L is sufficiently large), it is reasonable to assume that the quantization error Q is a *uniformly distributed random* variable, and the interfering effect of the quantization noise on the quantizer input is similar to that of thermal noise. We may thus express the probability density function of the quantization error Q as follows:

$$f_Q(q) = \begin{cases} 1/\Delta, & -\Delta/2 < q \leq \Delta/2 \\ 0, & \text{otherwise} \end{cases} \quad (\text{iv})$$

For this to be true, however, we must ensure that the incoming signal does not overload the quantizer. Then, with the mean of the quantization error being zero, its variance σ^2_Q is the same as the mean-square value:

$$\sigma^2_Q = E [Q^2] \quad (\text{v})$$

$$\begin{aligned} & \Delta/2 \\ &= \int_{-\Delta/2}^{\Delta/2} q^2 f_Q(q) dq \end{aligned}$$

Substituting equation (iv) and (v), we get

$$\begin{aligned} & \Delta/2 \\ \sigma^2_Q &= 1/\Delta \int_{-\Delta/2}^{\Delta/2} q^2 dq \quad (\text{vi}) \\ &= \Delta^2/12 \end{aligned}$$

Typically, the L -ary number k , denoting the k th representation level of the quantizer, is transmitted to the receiver in binary form. Let R denote the *bits per sample* used in the construction of the binary code. We may then write

$$L = 2^R \quad (\text{vii})$$

Or, equivalently,

$$R = \log_2 L \quad (\text{viii})$$

Hence, substituting equation (vii) into (iii), we get the step size

$$\Delta = 2 m_{\max}/2^R \quad (\text{ix})$$

Thus the use of equation (ix) in (viii) yields

$$\sigma^2 Q = 1/3 m_{\max}^2 2^{-2R} \quad (\text{x})$$

EQUIPMENTS:

- DCL-DPCM Kit
- Connecting Chords
- Power Supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

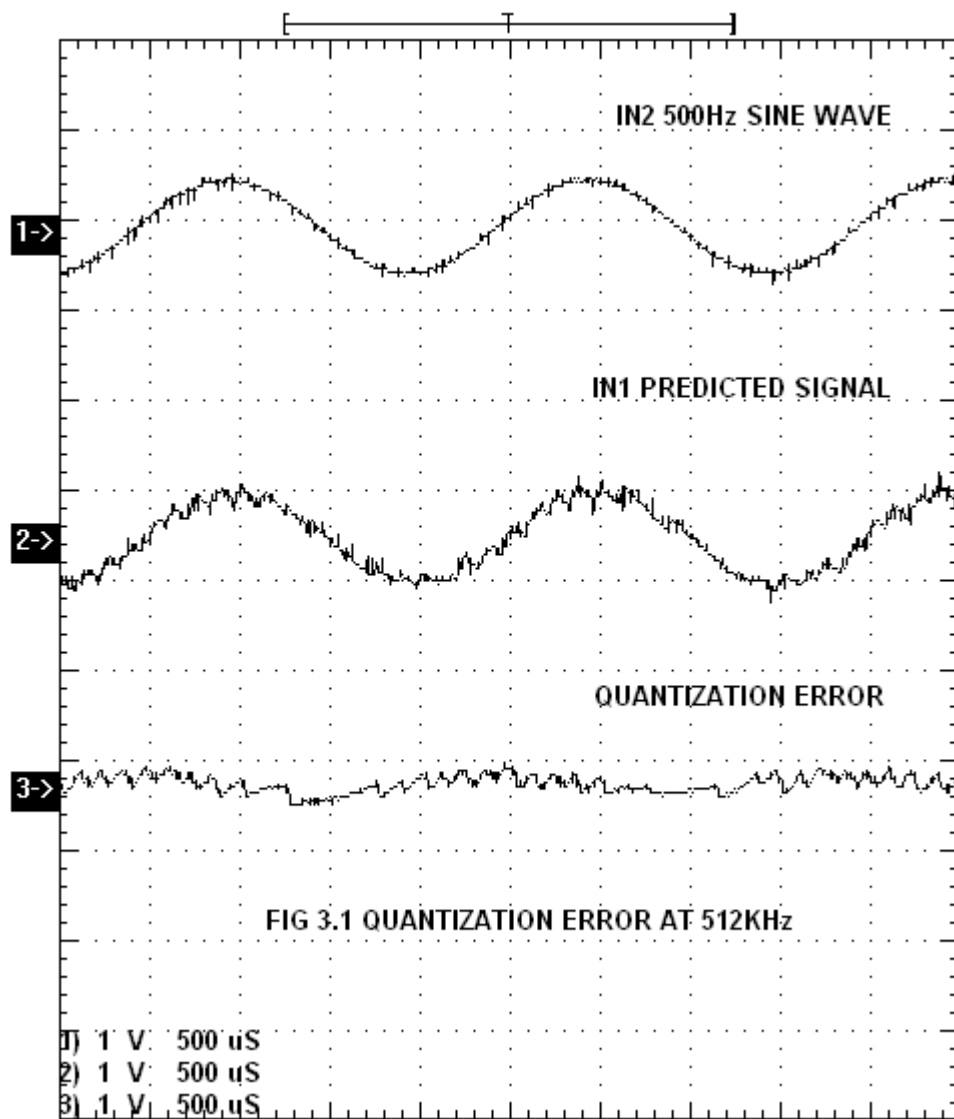
1. Refer to the block diagram (Fig.3) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it on.
3. Keep the input sine wave 500Hz to 1Vpp.
4. Select the clock rate to 512 KHz by changing the jumper position of JP1 in the clock generator section.
5. Connect the sine wave to the **IN** post of Analog Buffer.
6. Connect the **OUT** post of buffer to **IN** post of the DPCM modulator.
7. Also connect the **OUT** post of buffer to the **IN2** post of the quantization error block.
8. Connect the predicted signal out from the **PREDICTED OUT** post of the DPCM modulator section to the **IN1** post of the quantization error block.
9. Observe various waveforms as mentioned below.

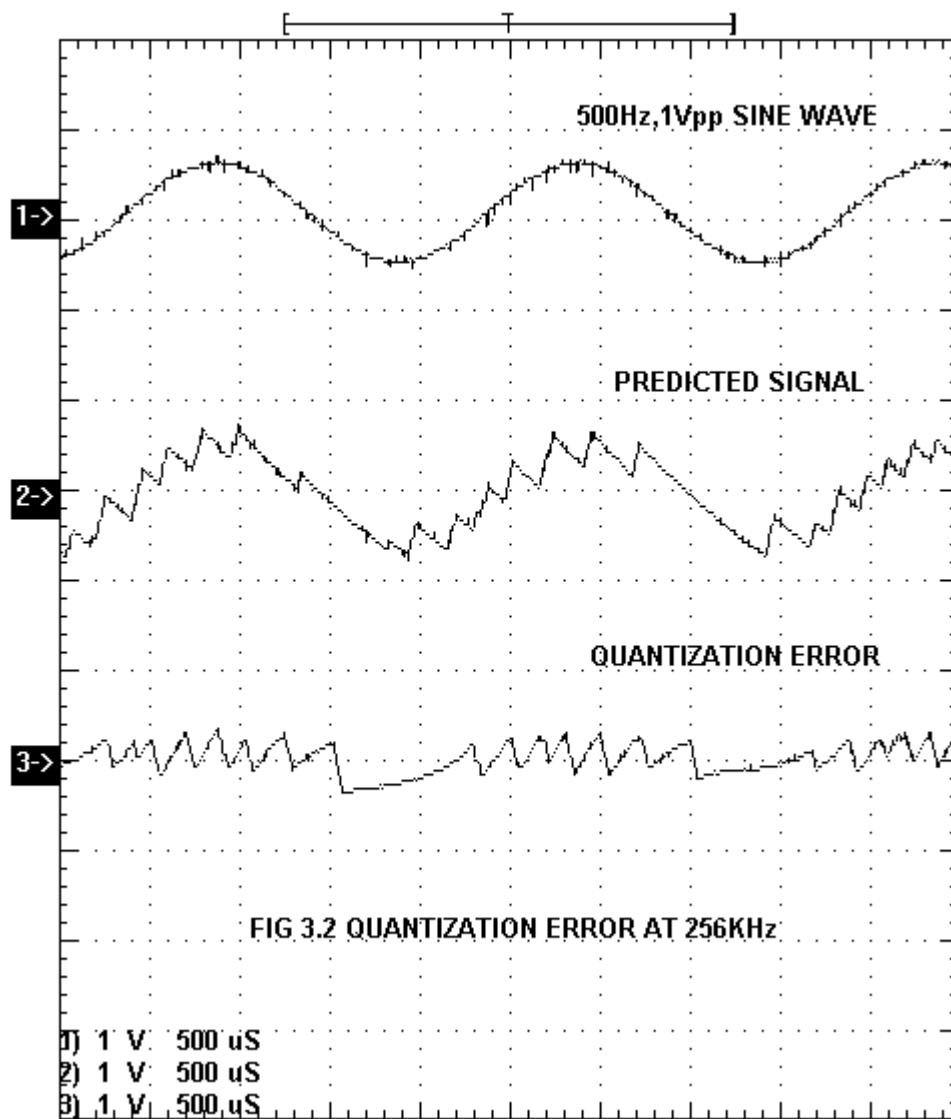
OBSERVATION:

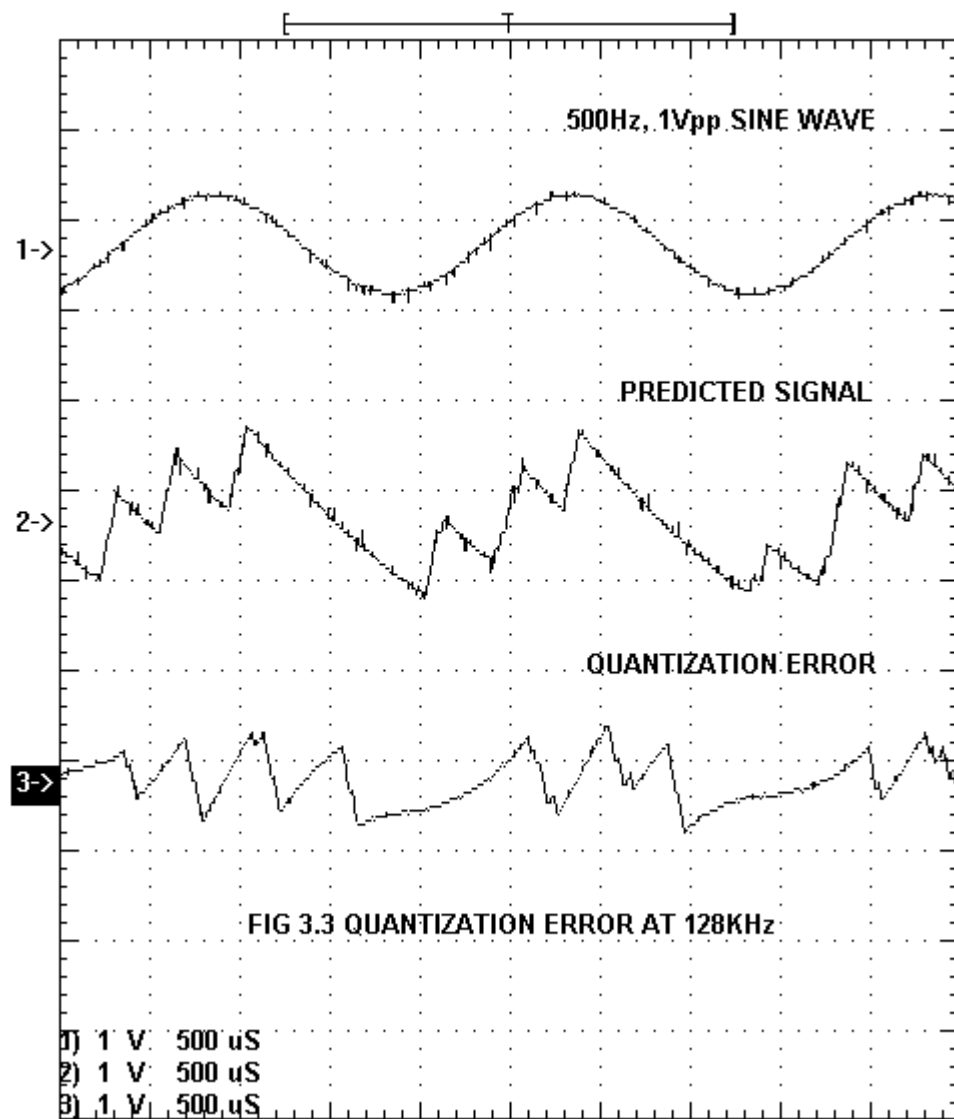
Observe the following waveforms on CRO and plot on the paper.

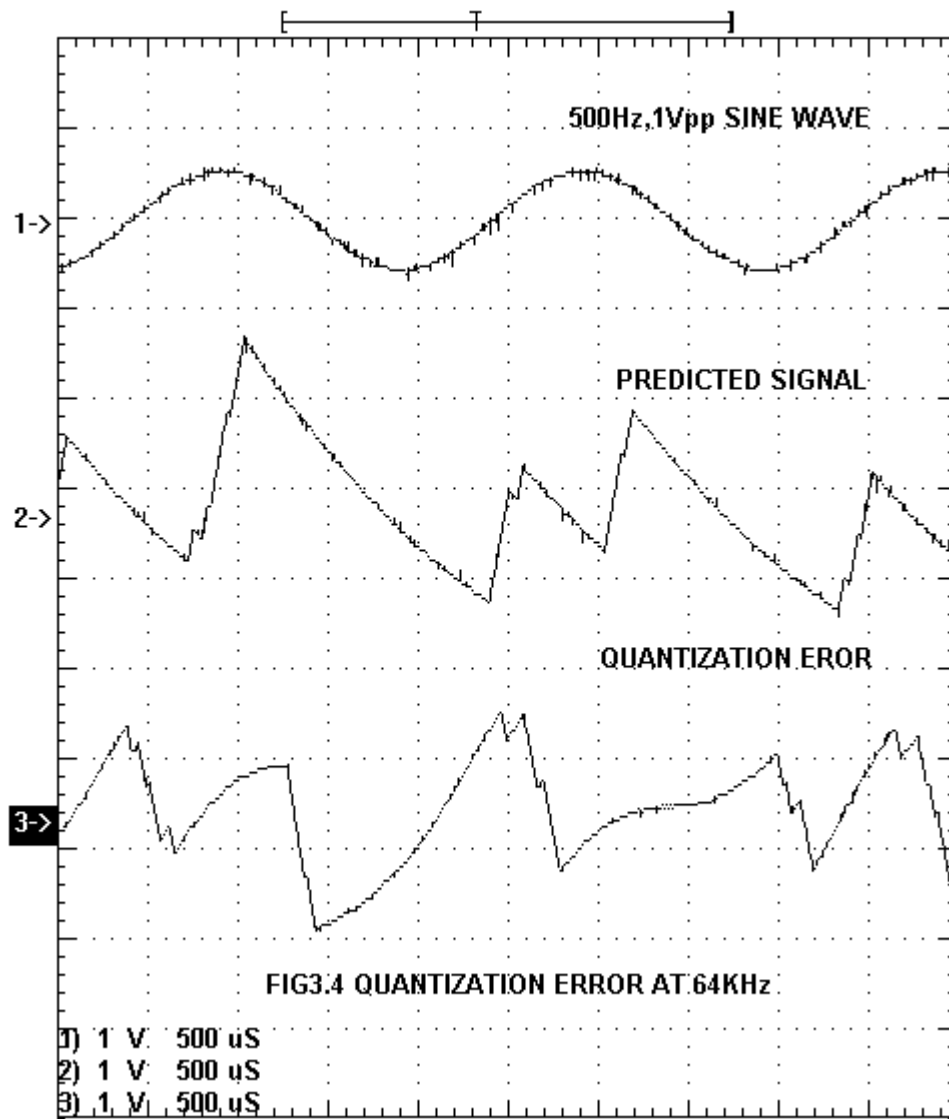
ON KIT DCL- DPCM

- Input sine wave, 500Hz, 1Vpp.
- Predicted signal at the **PREDICTED OUT** post of the DPCM modulator.
- Quantization error at the **OUT** post of the quantization error section.









**EXPERIMENT
NO. 4**

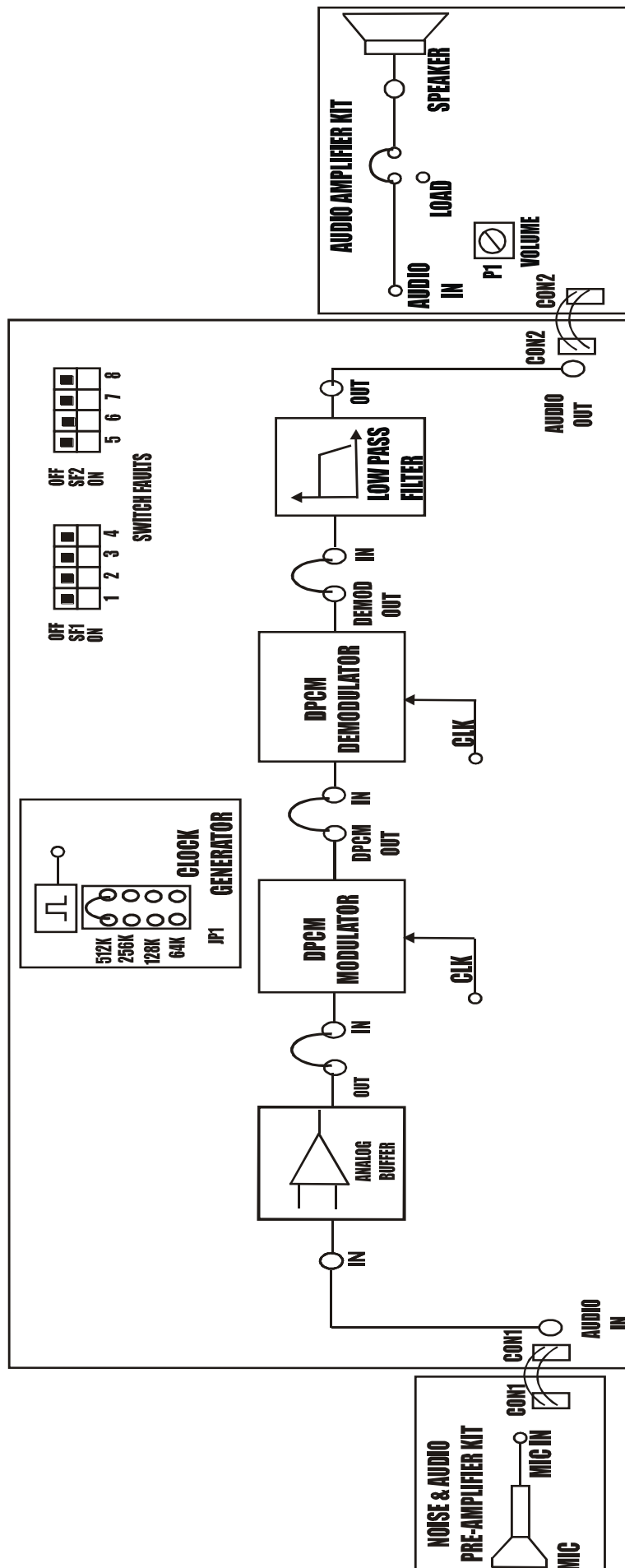


FIG. 4(a) BLOCK DIAGRAM FOR VOICE COMMUNICATION USING DPCM

EXPERIMENT NO: 4

NAME:

Voice Communication via DPCM/ADPCM

OBJECTIVE:

To study the voice communication using Differential Pulse Code Modulation and Adaptive Differential Pulse Code Modulation

THEORY:

Voice being the analog signal can be modulated using different methods, among this one method is the Differential Pulse Code Modulation. For performing the voice communication, set up the basic experiment mentioned earlier. DPCM reduces the bit rate for voice transmission. DPCM quantizes and encodes the difference between a previous sample input signal and a current sample input signal. DPCM quantizes the difference signal using uniform quantization. Since most of the signals generated by the human voice are small. Voice quality needs to focus on small signals. Therefore to solve this problem Adaptive Differential Pulse Code Modulation is developed.

EQUIPMENTS:

- DCL-DPCM Kit
- Noise and Audio Pre-Amplifier kit
- AI/O Kit (optional)
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE

A) Voice Communication via DPCM

1. Refer to the block diagram (Fig. 4[a]) and carry out the following connections and switch settings.
2. Keep the clock rate at 512 KHz, by changing the jumper position of JP1 in the clock generator section.
3. Connect the Noise and Audio Pre-Amplifier kit to the **CON1** post on **DCL-DPCM** kit using power connection cable.
4. Connect the Audio Amplifier kit to the **CON2** post on **DCL-DPCM** kit using power connection cable.

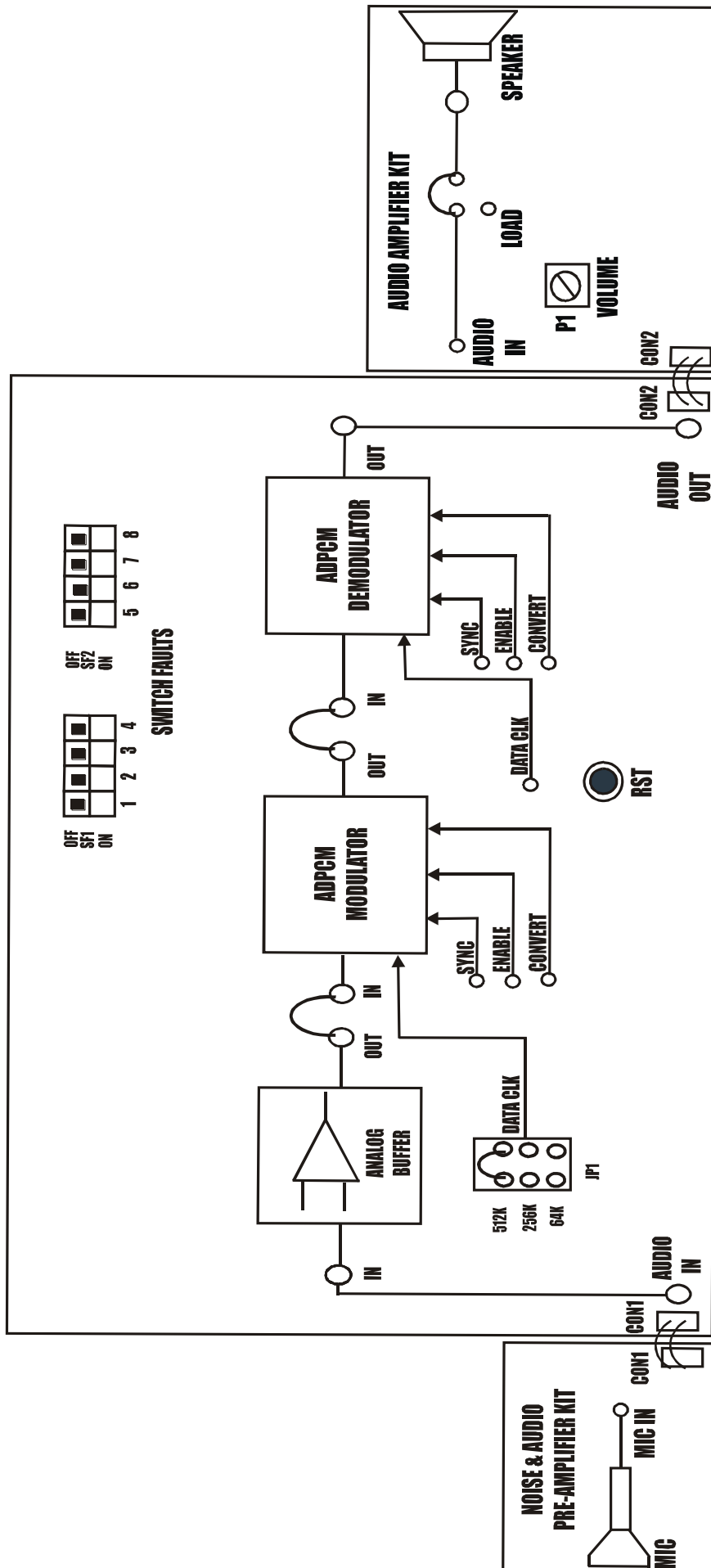


FIG. 4(b) BLOCK DIAGRAM FOR VOICE COMMUNICATION USING ADPCM

5. Connect Microphone to the Noise and Audio Pre-Amplifier kit at the **MIC IN** post.
6. Connect the input signal from **AUDIO IN** post to the **IN** post of the analog buffer.
7. Connect the patch cord from the **OUT** post of the analog buffer to the **IN** post of the DPCM modulator.
8. Connect the patch cord from the **DPCM OUT** post of the DPCM modulator to the **IN** post of the DPCM demodulator.
9. Connect the patch cord from the **DEMOD OUT** post of the DPCM demodulator to the **IN** post of the filter.
10. Connect the patch cord from the **OUT** post of the filter to the **AUDIO OUT** post of **ADCL-07**.
11. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
12. You can hear the voice signal at the output of the Audio Amplifier Kit. Use **RST** switch for clear reception of voice.
13. Repeat voice link experiment following the above procedure for different clock rates.

B) Voice Communication via ADPCM

PROCEDURE

1. Refer to the block diagram (Fig.4.1) and carry out the following connections and switch settings.
2. Keep the clock rate at 512 KHz, by changing the jumper position of JP1.
3. Connect the Noise and Audio Pre-Amplifier kit to the **CON1** post on **DCL-DPCM** kit using power connection cable.
4. Connect the Audio Amplifier kit to the **CON2** post on **DCL-DPCM** kit using power connection cable.
5. Connect Microphone to the Noise and Audio Pre-Amplifier kit at **MIC IN** post.
6. Connect the input signal from **AUDIO IN** post to the **IN** post of the analog buffer.
7. Connect the patch cord from the **OUT** post of the analog buffer to the **IN** post of the ADPCM modulator.
8. Connect the patch cord from the **OUT** post of the ADPCM modulator to the **IN** post of the ADPCM demodulator.
9. Connect the patch cord from the **OUT** post of the ADPCM demodulator to the **AUDIO OUT**.
10. Connect power supply in proper polarity to the kit **DCL-DPCM** and switch it ON.
11. You can hear the voice signal at the output of the Audio Amplifier Kit. Use **RST** switch for clear reception of voice.
12. Repeat voice link experiment following the above procedure for different clock rates.